MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

49PR FUEL TREATMENT HOUSE

FOUNDATION DESIGN CALCULATION

AS BUILT

MD-216-49PR-EG-CV-CAL-1635

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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01
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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2018	Saudi Concrete Structures Code
ACI 318 :2014	Building Code Requirement of Structural Concrete

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document
LRFD	Load and Resistance Factor Design
F.O.S	Factor of Safety

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1. INTENT

The document is prepared for Design Calculation of Column Foundation for **FUEL TREATMENT HOUSE**

2. SCOPE

The document is prepared for Design Calculation of Column Foundation for FUEL TREATMENT HOUSE
All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
The foundation is designed for wind load as seismic load is less than wind load.(Refere MD-216-49PR-EG-ST-CAL-2407 for Fuel treatment structural design calculation)
Design of Reinforced Concrete Foundations followed Strength Design as per SBC 304:2018 & ACI 318:2008
Net Safe Bearing Capaciity is obtained from Geotechnical report BH28 . Approximate value of 50T/sq.m is considered

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3. Design of FUEL TREATMENT HOUSE

3.1 Design of Foundation mkd. F1 & F1A

Load Detail	A3 TO A5, C3 TO C5		
	FX (T)	FY (T)	FZ (T)
1 DL	-1	7	0
2 LLROOF	-1	4	0
3 CL1(L+XS)	0	4	0
4 CL2(L-XS)	-1	4	0
5 CL3(R+XS)	0	2	0
6 CL4(R-XS)	-1	2	0
7 CL5(L+ZT)	0	4	0
8 CL6(L-ZT)	0	4	0
9 CL7(R+ZT)	0	2	0
10 CL8(R-ZT)	0	2	0
21 WL1(+X)S	2	-2	0
22 WL2(+X)P	2	-3	0
23 WL3(-X)S	0	-1	0
24 WL4(-X)P	-1	-1	0
25 WL5(+Z)S	0	-2	1
26 WL6(+Z)P	0	-3	-1
27 WL7(-Z)S	0	-2	1
28 WL8(-Z)P	0	-3	-1

STRENGTH DESIGN	A3 TO A5, C3 TO C5		
Load Combinations	FX (T)	FY (T)	FZ (T)
1.4D	-1.58	9.44	0.00
1.4D+1.7(CL1)+0.5Lr	-2.5	17.5	0.0
1.4D+1.7(CL2)+0.5Lr	-3.2	18.1	0.0
1.4D+1.7(CL3)+0.5Lr	-2.5	14.7	0.0
1.4D+1.7(CL4)+0.5Lr	-3.1	15.3	0.0
1.4D+1.7(CL5)+0.5Lr	-2.8	17.8	0.4
1.4D+1.7(CL6)+0.5Lr	-2.8	17.8	-0.4
1.4D+1.7(CL5)+0.5Lr	-2.8	15.0	0.4
1.4D+1.7(CL6)+0.5Lr	-2.5	14.7	-0.4
1.2D+1.6Lr+1.0(CL1)	-3.4	18.1	0.0
1.2D+1.6Lr+1.0(CL2)	-3.8	18.5	0.0
1.2D+1.6Lr+1.0(CL3)	-3.4	16.5	0.0
1.2D+1.6Lr+1.0(CL4)	-3.8	16.8	0.0
1.2D+1.6Lr+1.0(CL5)	-3.6	18.3	0.3
1.2D+1.6Lr+1.0(CL6)	-3.6	18.3	-0.3
1.2D+1.6Lr+1.0(CL7)	-3.6	16.7	0.3
1.2D+1.6Lr+1.0(CL8)	-3.4	16.4	-0.3
1.2D + 1.6WL1+1.0L+0.5Lr	1.2	6.5	0.0
1.2D + 1.6WL2+1.0L+0.5Lr	0.8	4.9	0.0
1.2D + 1.6WL3+1.0L+0.5Lr	-2.5	9.1	0.0
1.2D + 1.6WL4+1.0L+0.5Lr	-3.0	7.8	0.0
1.2D + 1.6WL5+1.0L+0.5Lr	-1.8	7.3	1.9
1.2D + 1.6WL6+1.0L+0.5Lr	-2.1	5.3	-1.9
1.2D + 1.6WL7+1.0L+0.5Lr	-2.0	7.6	1.9
1.2D + 1.6WL8+1.0L+0.5Lr	-2.1	5.4	-1.9
0.9D+1.6WL1	2.1	2.5	0.0
0.9D+1.6WL2	1.7	0.9	0.0
0.9D+1.6WL3	-1.6	5.0	0.0
0.9D+1.6WL4	-2.1	3.8	0.0
0.9D+1.6WL5	-0.9	3.2	1.9
0.9D+1.6WL6	-1.1	1.2	-1.9
0.9D+1.6WL7	-1.1	3.5	1.9
0.9D+1.6WL8	-1.2	1.4	-1.9

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DESIGN OF FOUNDATION MKD. F1

	About-X	About-Z
Size of column :	0.55m	0.55m
Size of fdn. raft :	1.80m	1.80m
Thickness of raft :	0.30 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	0.50 m	
Unit weight of conc :	2.5 T/m^3	
Unit weight of soil :	1.8 T/m^3	

SYMBOLS AND NOTATIONS

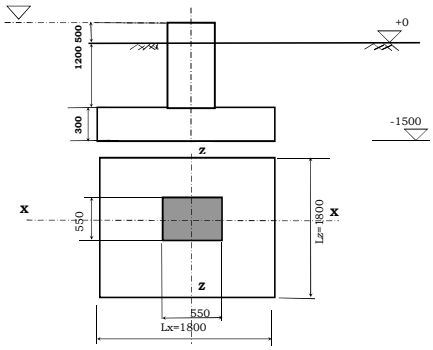
Dead Load	D
live load	LL
Roof live load	Lr
Wind Load	W
Crane Load	CL

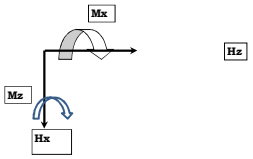
Calculation of Weight of Foundation :

Weight of raft & Col.= (1.8*1.8*0.3*2.5)+(0.55*0.55*1.7*2.5) = 3.72T

Weight of soil = ((1.8*1.8)-(0.55*0.55))*1.2*1.8 = 6.35T

U/S Base Pl Total = 10.06T





Design Input:

Characteristic strength of concrete, f'$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 217mm = 0.217m

Effective depth of slab along Z direction = 201mm = 0.201m

Clear projection along Z direction = 0.625m

Clear projection along X direction = 0.625m

WORKING STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	M_X	M_Z	V (T)	Hx	MX	Hz	MZ
DL+Lr	-2.28	10.79	0.00	0.0	0.00	20.85	-2.28	0.00	0.00	4.56
DL+CL1	-1.31	10.29	0.00	0.0	0.00	20.35	-1.31	0.00	0.00	2.61
DL+CL2	-1.72	10.65	0.00	0.0	0.00	20.71	-1.72	0.00	0.00	3.43
DL+CL3	-1.33	8.67	0.00	0.0	0.00	18.73	-1.33	0.00	0.00	2.65
DL+CL4	-1.70	9.02	0.00	0.0	0.00	19.08	-1.70	0.00	0.00	3.39
DL+CL5	-1.51	10.47	0.26	0.0	0.00	20.53	-1.51	0.52	0.26	3.02
DL+CL6	-1.51	10.48	-0.26	0.0	0.00	20.54	-1.51	0.52	-0.26	3.03
DL+CL5	-1.51	8.84	0.26	0.0	0.00	18.90	-1.51	0.52	0.26	3.02
DL+CL6	-1.31	8.62	-0.26	0.0	0.00	18.68	-1.31	0.52	-0.26	2.63
DL+0.75*CL1+0.75*Lr	-2.12	12.44	0.00	0.0	0.00	22.50	-2.12	0.00	0.00	4.25
DL+0.75*CL2+0.75*Lr	-2.43	12.71	0.00	0.0	0.00	22.77	-2.43	0.00	0.00	4.86
DL+0.75*CL3+0.75*Lr	-2.14	11.23	0.00	0.0	0.00	21.29	-2.14	0.00	0.00	4.28
DL+0.75*CL4+0.75*Lr	-2.42	11.48	0.00	0.0	0.00	21.54	-2.42	0.00	0.00	4.83
DL+0.75*CL5+0.75*Lr	-2.28	12.57	0.20	0.0	0.00	22.63	-2.28	0.39	0.20	4.55
DL+0.75*CL6+0.75*Lr	-2.28	12.58	-0.20	0.0	0.00	22.64	-2.28	0.39	-0.20	4.56
DL+0.75*CL7+0.75*Lr	-2.28	11.35	0.20	0.0	0.00	21.41	-2.28	0.39	0.20	4.55
DL+0.75*CL8+0.75*Lr	-2.13	11.18	-0.20	0.0	0.00	21.24	-2.13	0.39	-0.20	4.26
DL+0.75*Lr+0.75*WL1	-0.53	8.10	0.00	0.0	0.00	18.16	-0.53	0.00	0.00	1.06
DL+0.75*Lr+0.75*WL2	-0.70	7.35	0.00	0.0	0.00	17.41	-0.70	0.00	0.00	1.41
DL+0.75*Lr+0.75*WL3	-2.28	9.30	0.00	0.0	0.00	19.36	-2.28	0.00	0.00	4.56
DL+0.75*Lr+0.75*WL4	-2.49	8.72	0.00	0.0	0.00	18.78	-2.49	0.00	0.00	4.98
DL+0.75*Lr+0.75*WL5	-1.92	8.44	0.88	0.0	0.00	18.50	-1.92	1.76	0.88	3.84
DL+0.75*Lr+0.75*WL6	-2.05	7.52	-0.88	0.0	0.00	17.58	-2.05	1.76	-0.88	4.10
DL+0.75*Lr+0.75*WL7	-2.02	8.58	0.88	0.0	0.00	18.64	-2.02	1.76	0.88	4.05
DL+0.75*Lr+0.75*WL8	-2.08	7.57	-0.88	0.0	0.00	17.63	-2.08	1.76	-0.88	4.16
0.6*DL+WL1	1.27	1.81	0.00	0.0	0.00	11.87	1.27	0.00	0.00	2.54
0.6*DL+WL2	1.04	0.81	0.00	0.0	0.00	10.87	1.04	0.00	0.00	2.08
0.6*DL+WL3	-1.06	3.40	0.00	0.0	0.00	13.46	-1.06	0.00	0.00	2.12
0.6*DL+WL4	-1.34	2.63	0.00	0.0	0.00	12.69	-1.34	0.00	0.00	2.69
0.6*DL+WL5	-0.59	2.26	1.17	0.0	0.00	12.32	-0.59	2.34	1.17	1.17
0.6*DL+WL6	-0.76	1.03	-1.17	0.0	0.00	11.09	-0.76	2.34	-1.17	1.52
0.6*DL+WL7	-0.72	2.45	1.17	0.0	0.00	12.51	-0.72	2.34	1.17	1.45
0.6*DL+WL8	-0.79	1.11	-1.17	0.0	0.00	11.17	-0.79	2.34	-1.17	1.59

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CHECK FOR OVERTURNING AND SLIDING

	Load at Foundation Base					Overturning & Sliding Check (Coeffecient, μ = 0.4)			
	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
Load Case									
DL+Lr	20.85	-2.28	4.56	0.00	0.00	4.12	NA	3.66	NA
DL+CL1	20.35	-1.31	2.61	0.00	0.00	7.02	NA	6.24	NA
DL+CL2	20.71	-1.72	3.43	0.00	0.00	5.43	NA	4.83	NA
DL+CL3	18.73	-1.33	2.65	0.00	0.00	6.36	NA	5.65	NA
DL+CL4	19.08	-1.70	3.39	0.00	0.00	5.06	NA	4.50	NA
DL+CL5	20.53	-1.51	3.02	0.26	0.52	6.12	35.52	5.44	31.58
DL+CL6	20.54	-1.51	3.03	-0.26	0.52	6.11	35.54	5.43	31.59
DL+CL5	18.90	-1.51	3.02	0.26	0.52	5.64	32.71	5.01	29.08
DL+CL6	18.68	-1.31	2.63	-0.26	0.52	6.39	32.32	5.68	28.73
DL+0.75*CL1+0.75*Lr	22.50	-2.12	4.25	0.00	0.00	4.77	NA	4.24	NA
DL+0.75*CL2+0.75*Lr	22.77	-2.43	4.86	0.00	0.00	4.21	NA	3.75	NA
DL+0.75*CL3+0.75*Lr	21.29	-2.14	4.28	0.00	0.00	4.48	NA	3.98	NA
DL+0.75*CL4+0.75*Lr	21.54	-2.42	4.83	0.00	0.00	4.01	NA	3.57	NA
DL+0.75*CL5+0.75*Lr	22.63	-2.28	4.55	0.20	0.39	4.48	52.23	3.98	46.42
DL+0.75*CL6+0.75*Lr	22.64	-2.28	4.56	-0.20	0.39	4.47	52.24	3.97	46.44
DL+0.75*CL7+0.75*Lr	21.41	-2.28	4.55	0.20	0.39	4.24	49.41	3.76	43.92
DL+0.75*CL8+0.75*Lr	21.24	-2.13	4.26	-0.20	0.39	4.49	49.03	3.99	43.58
DL+0.75*Lr+0.75*WL1	18.16	-0.53	1.06	0.00	0.00	15.43	NA	13.71	NA
DL+0.75*Lr+0.75*WL2	17.41	-0.70	1.41	0.00	0.00	11.14	NA	9.90	NA
DL+0.75*Lr+0.75*WL3	19.36	-2.28	4.56	0.00	0.00	3.82	NA	3.40	NA
DL+0.75*Lr+0.75*WL4	18.78	-2.49	4.98	0.00	0.00	3.39	NA	3.02	NA
DL+0.75*Lr+0.75*WL5	18.50	-1.92	3.84	0.88	1.76	4.33	9.49	3.85	8.43
DL+0.75*Lr+0.75*WL6	17.58	-2.05	4.10	-0.88	1.76	3.86	9.01	3.43	8.01
DL+0.75*Lr+0.75*WL7	18.64	-2.02	4.05	0.88	1.76	4.14	9.56	3.68	8.50
DL+0.75*Lr+0.75*WL8	17.63	-2.08	4.16	-0.88	1.76	3.82	9.04	3.39	8.04
0.6*DL+WL1	11.87	1.27	2.54	0.00	0.00	4.20	NA	3.74	NA
0.6*DL+WL2	10.87	1.04	2.08	0.00	0.00	4.71	NA	4.19	NA
0.6*DL+WL3	13.46	-1.06	2.12	0.00	0.00	5.71	NA	5.08	NA
0.6*DL+WL4	12.69	-1.34	2.69	0.00	0.00	4.25	NA	3.78	NA
0.6*DL+WL5	12.32	-0.59	1.17	1.17	2.34	9.46	4.74	8.41	4.21
0.6*DL+WL6	11.09	-0.76	1.52	-1.17	2.34	6.58	4.27	5.84	3.79
0.6*DL+WL7	12.51	-0.72	1.45	1.17	2.34	7.79	4.81	6.92	4.28
0.6*DL+WL8	11.17	-0.79	1.59	-1.17	2.34	6.33	4.29	5.63	3.82

ULTIMATE STRESS DESIGN	**Load at U/S Of Base Plt**					**Load at bottom of foundation**		
Load Case	FX (T)	FY(T)	FZ(T)	M_x	M_z	V (T)	Mx (T-m)	Mz (T-m)
1.4D	-1.58	9.44	0.00	0.0	0.00	19.497	0.00	3.16
1.4D+1.7(CL1)+0.5Lr	-2.45	17.50	0.00	0.0	0.00	27.564	0.00	4.91
1.4D+1.7(CL2)+0.5Lr	-3.15	18.10	0.00	0.0	0.00	28.160	0.00	6.31
1.4D+1.7(CL3)+0.5Lr	-2.49	14.75	0.00	0.0	0.00	24.810	0.00	4.98
1.4D+1.7(CL4)+0.5Lr	-3.12	15.33	0.00	0.0	0.00	25.389	0.00	6.24
1.4D+1.7(CL5)+0.5Lr	-2.80	17.79	0.44	0.0	0.00	27.853	0.88	5.60
1.4D+1.7(CL6)+0.5Lr	-2.81	17.81	-0.44	0.0	0.00	27.871	0.88	5.62
1.4D+1.7(CL5)+0.5Lr	-2.80	15.03	0.44	0.0	0.00	25.091	0.88	5.60
1.4D+1.7(CL6)+0.5Lr	-2.47	14.65	-0.44	0.0	0.00	24.710	0.88	4.94
1.2D+1.6Lr+1.0(CL1)	-3.37	18.12	0.00	0.0	0.00	28.182	0.00	6.74
1.2D+1.6Lr+1.0(CL2)	-3.78	18.47	0.00	0.0	0.00	28.533	0.00	7.56
1.2D+1.6Lr+1.0(CL3)	-3.39	16.50	0.00	0.0	0.00	26.562	0.00	6.78
1.2D+1.6Lr+1.0(CL4)	-3.76	16.84	0.00	0.0	0.00	26.903	0.00	7.52
1.2D+1.6Lr+1.0(CL5)	-3.57	18.29	0.26	0.0	0.00	28.352	0.52	7.14
1.2D+1.6Lr+1.0(CL6)	-3.58	18.30	-0.26	0.0	0.00	28.363	0.52	7.16
1.2D+1.6Lr+1.0(CL7)	-3.57	16.67	0.26	0.0	0.00	26.727	0.52	7.14
1.2D+1.6Lr+1.0(CL8)	-3.38	16.44	-0.26	0.0	0.00	26.504	0.52	6.76
1.2D + 1.6WL1+1.0L+0.5Lr	1.19	6.53	0.00	0.0	0.00	16.594	0.00	2.37
1.2D + 1.6WL2+1.0L+0.5Lr	0.82	4.94	0.00	0.0	0.00	15.002	0.00	1.63
1.2D + 1.6WL3+1.0L+0.5Lr	-2.54	9.08	0.00	0.0	0.00	19.144	0.00	5.08
1.2D + 1.6WL4+1.0L+0.5Lr	-3.00	7.85	0.00	0.0	0.00	17.908	0.00	5.99
1.2D + 1.6WL5+1.0L+0.5Lr	-1.78	7.25	1.87	0.0	0.00	17.313	3.74	3.57
1.2D + 1.6WL6+1.0L+0.5Lr	-2.06	5.29	-1.87	0.0	0.00	15.350	3.74	4.12
1.2D + 1.6WL7+1.0L+0.5Lr	-2.00	7.55	1.87	0.0	0.00	17.615	3.74	4.00
1.2D + 1.6WL8+1.0L+0.5Lr	-2.12	5.41	-1.87	0.0	0.00	15.473	3.74	4.23
0.9D+1.6WL1	2.10	2.49	0.00	0.0	0.00	12.547	0.00	4.20
0.9D+1.6WL2	1.73	0.90	0.00	0.0	0.00	10.956	0.00	3.46
0.9D+1.6WL3	-1.63	5.04	0.00	0.0	0.00	15.097	0.00	3.26
0.9D+1.6WL4	-2.08	3.80	0.00	0.0	0.00	13.861	0.00	4.17
0.9D+1.6WL5	-0.87	3.21	1.87	0.0	0.00	13.266	3.74	1.74
0.9D+1.6WL6	-1.15	1.24	-1.87	0.0	0.00	11.303	3.74	2.29
0.9D+1.6WL7	-1.09	3.51	1.87	0.0	0.00	13.568	3.74	2.18
0.9D+1.6WL8	-1.20	1.37	-1.87	0.0	0.00	11.426	3.74	2.41

Check for Base Pressure

Load Combination & Calculation of Base pressure

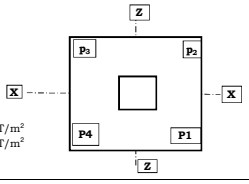
Max. Base pressure = p_{max} = $(V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = p_{min} = $(V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Appox. Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 (BH28) For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 53.6 T/m^2 For WL/SL case = 66.1 T/m^2



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Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	05-06-2020	08-07-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
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Prepared By	SHD	SHD	SHD
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Load Case	Design Load/moments V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	Base Pressure (T/m^2) p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	Remarks
DL+Lr	20.85	4.56	0.00	6.44	4.69	0.00	13.12	13.12	3.75	3.75	10.96	O.K
DL+CL1	20.35	2.61	0.00	6.28	2.69	0.00	10.97	10.97	5.60	5.60	8.81	O.K
DL+CL2	20.71	3.43	0.00	6.39	3.53	0.00	11.92	11.92	4.86	4.86	9.76	O.K
DL+CL3	18.73	2.65	0.00	5.78	2.73	0.00	10.51	10.51	5.05	5.05	8.35	O.K
DL+CL4	19.08	3.39	0.00	5.89	3.49	0.00	11.38	11.38	4.40	4.40	9.22	O.K
DL+CL5	20.53	3.02	0.52	6.33	3.10	0.53	11.97	10.90	5.77	4.70	9.81	O.K
DL+CL6	20.54	3.03	0.52	6.34	3.11	0.53	11.99	10.92	5.76	4.69	9.83	O.K
DL+CL5	18.90	3.02	0.52	5.83	3.10	0.53	11.47	10.40	5.27	4.20	9.31	O.K
DL+CL6	18.68	2.63	0.52	5.76	2.70	0.53	11.00	9.93	5.59	4.53	8.84	O.K
DL+0.75*CL1+0.75*Lr	22.50	4.25	0.00	6.95	4.37	0.00	13.31	13.31	4.58	4.58	11.15	O.K
DL+0.75*CL2+0.75*Lr	22.77	4.86	0.00	7.03	5.00	0.00	14.03	14.03	4.02	4.02	11.87	O.K
DL+0.75*CL3+0.75*Lr	21.29	4.28	0.00	6.57	4.40	0.00	12.97	12.97	4.17	4.17	10.81	O.K
DL+0.75*CL4+0.75*Lr	21.54	4.83	0.00	6.65	4.97	0.00	13.62	13.62	3.68	3.68	11.46	O.K
DL+0.75*CL5+0.75*Lr	22.63	4.55	0.39	6.98	4.68	0.40	14.07	13.26	4.71	3.90	11.91	O.K
DL+0.75*CL6+0.75*Lr	22.64	4.56	0.39	6.99	4.69	0.40	14.08	13.28	4.70	3.90	11.92	O.K
DL+0.75*CL7+0.75*Lr	21.41	4.55	0.39	6.61	4.68	0.40	13.69	12.89	4.33	3.53	11.53	O.K
DL+0.75*CL8+0.75*Lr	21.24	4.26	0.39	6.56	4.38	0.40	13.34	12.54	4.58	3.77	11.18	O.K
DL+0.75*Lr+0.75*WL1	18.16	1.06	0.00	5.60	1.09	0.00	8.69	8.69	6.51	6.51	6.53	O.K
DL+0.75*Lr+0.75*WL2	17.41	1.41	0.00	5.37	1.45	0.00	8.82	8.82	5.93	5.93	6.66	O.K
DL+0.75*Lr+0.75*WL3	19.36	4.56	0.00	5.97	4.69	0.00	12.66	12.66	3.29	3.29	10.50	O.K
DL+0.75*Lr+0.75*WL4	18.78	4.98	0.00	5.80	5.12	0.00	12.92	12.92	2.67	2.67	10.76	O.K
DL+0.75*Lr+0.75*WL5	18.50	3.84	1.76	5.71	3.95	1.81	13.47	9.86	5.56	1.95	11.31	O.K
DL+0.75*Lr+0.75*WL6	17.58	4.10	1.76	5.42	4.22	1.81	13.45	9.84	5.01	1.40	11.29	O.K
DL+0.75*Lr+0.75*WL7	18.64	4.05	1.76	5.75	4.17	1.81	13.72	10.11	5.39	1.78	11.56	O.K
DL+0.75*Lr+0.75*WL8	17.63	4.16	1.76	5.44	4.28	1.81	13.52	9.91	4.97	1.36	11.36	O.K
0.6*DL+WL1	11.87	2.54	0.00	3.66	2.61	0.00	8.28	8.28	3.05	3.05	6.12	O.K
0.6*DL+WL2	10.87	2.08	0.00	3.36	2.14	0.00	7.49	7.49	3.22	3.22	5.33	O.K
0.6*DL+WL3	13.46	2.12	0.00	4.15	2.18	0.00	8.34	8.34	3.97	3.97	6.18	O.K
0.6*DL+WL4	12.69	2.69	0.00	3.92	2.77	0.00	8.68	8.68	3.15	3.15	6.52	O.K
0.6*DL+WL5	12.32	1.17	2.34	3.80	1.21	2.41	9.41	4.60	7.00	2.19	7.25	O.K
0.6*DL+WL6	11.09	1.52	2.34	3.42	1.56	2.41	9.39	4.58	6.27	1.45	7.23	O.K
0.6*DL+WL7	12.51	1.45	2.34	3.86	1.49	2.41	9.75	4.94	6.78	1.97	7.59	O.K
0.6*DL+WL8	11.17	1.59	2.34	3.45	1.63	2.41	9.49	4.67	6.22	1.41	7.33	O.K

Calculation of Design Base Pressure

Load Combination & Calculation of Base pressure

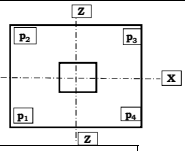
Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 53.6 T/m^2 For WL/SL case = 66.1 T/m^2



Load Case	Design Load/moments V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)
1.4D	19.5	3.16	0.00	6.02	3.25	0.00	11.27	11.27	4.77	4.77	8.57
1.4D+1.7(CL1)+0.5Lr	27.6	4.91	0.00	8.51	5.05	0.00	15.56	13.56	3.46	3.46	12.86
1.4D+1.7(CL2)+0.5Lr	28.2	6.31	0.00	8.69	6.49	0.00	17.18	15.18	2.20	2.20	14.48
1.4D+1.7(CL3)+0.5Lr	24.8	4.98	0.00	7.66	5.12	0.00	14.78	12.78	2.53	2.53	12.08
1.4D+1.7(CL4)+0.5Lr	25.4	6.24	0.00	7.84	6.42	0.00	16.25	14.25	1.42	1.42	13.55
1.4D+1.7(CL5)+0.5Lr	27.9	5.60	0.88	8.60	5.76	0.91	17.27	13.45	3.75	1.93	14.57
1.4D+1.7(CL6)+0.5Lr	27.9	5.62	0.88	8.60	5.78	0.91	17.29	13.47	3.73	1.91	14.59
1.4D+1.7(CL5)+0.5Lr	25.1	5.60	0.88	7.74	5.76	0.91	16.41	12.60	2.89	1.07	13.71
1.4D+1.7(CL6)+0.5Lr	24.7	4.94	0.88	7.63	5.08	0.91	15.62	11.80	3.45	1.63	12.92
1.2D+1.6Lr+1.0(CL1)	28.2	6.74	0.00	8.70	6.93	0.00	17.63	15.63	1.76	1.76	14.93
1.2D+1.6Lr+1.0(CL2)	28.5	7.56	0.00	8.81	7.78	0.00	18.59	16.59	1.03	1.03	15.89
1.2D+1.6Lr+1.0(CL3)	26.6	6.78	0.00	8.20	6.98	0.00	17.17	15.17	1.22	1.22	14.47
1.2D+1.6Lr+1.0(CL4)	26.9	7.52	0.00	8.30	7.74	0.00	18.04	16.04	0.57	0.57	15.34
1.2D+1.6Lr+1.0(CL5)	28.4	7.14	0.52	8.75	7.35	0.53	18.64	15.57	1.94	0.87	15.94
1.2D+1.6Lr+1.0(CL6)	28.4	7.16	0.52	8.75	7.36	0.53	18.65	15.58	1.93	0.86	15.95
1.2D+1.6Lr+1.0(CL7)	26.7	7.14	0.52	8.25	7.35	0.53	18.13	15.06	1.43	0.36	15.43
1.2D+1.6Lr+1.0(CL8)	26.5	6.76	0.52	8.18	6.95	0.53	17.67	14.60	1.76	0.69	14.97
1.2D + 1.6WL1+1.0L+0.5Lr	16.6	2.37	0.00	5.12	2.44	0.00	9.56	7.56	2.68	2.68	6.86
1.2D + 1.6WL2+1.0L+0.5Lr	15.0	1.63	0.00	4.63	1.68	0.00	8.31	6.31	2.95	2.95	5.61
1.2D + 1.6WL3+1.0L+0.5Lr	19.1	5.08	0.00	5.91	5.23	0.00	13.14	11.14	0.68	0.68	10.44
1.2D + 1.6WL4+1.0L+0.5Lr	17.9	5.99	0.00	5.53	6.16	0.00	13.69	11.69	-0.64	-0.64	10.99
1.2D + 1.6WL5+1.0L+0.5Lr	17.3	3.57	3.74	5.34	3.67	3.85	14.86	5.16	5.53	-2.18	12.16
1.2D + 1.6WL6+1.0L+0.5Lr	15.3	4.12	3.74	4.74	4.24	3.85	14.83	5.12	4.35	-3.35	12.13
1.2D + 1.6WL7+1.0L+0.5Lr	17.6	4.00	3.74	5.44	4.12	3.85	15.41	5.70	5.17	-2.53	12.71
1.2D + 1.6WL8+1.0L+0.5Lr	15.5	4.23	3.74	4.78	4.35	3.85	14.98	5.28	4.27	-3.43	12.28
0.9D+1.6WL1	12.5	4.20	0.00	3.87	4.32	0.00	10.19	8.19	-0.45	-0.45	7.49
0.9D+1.6WL2	11.0	3.46	0.00	3.38	3.56	0.00	8.94	6.94	-0.18	-0.18	6.24
0.9D+1.6WL3	15.1	3.26	0.00	4.66	3.35	0.00	10.01	8.01	1.31	1.31	7.31
0.9D+1.6WL4	13.9	4.17	0.00	4.28	4.29	0.00	10.56	8.56	-0.01	-0.01	7.86
0.9D+1.6WL5	13.3	1.74	3.74	4.09	1.79	3.85	11.74	2.03	6.16	-1.55	9.04
0.9D+1.6WL6	11.3	2.29	3.74	3.49	2.36	3.85	11.70	2.00	4.98	-2.72	9.00
0.9D+1.6WL7	13.6	2.18	3.74	4.19	2.24	3.85	12.28	2.58	5.80	-1.90	9.58
0.9D+1.6WL8	11.4	2.41	3.74	3.53	2.47	3.85	11.85	2.15	4.90	-2.80	9.15

Calculation of Bending Moment and Shear Force in X direction :

BM_{1x} =Moment at the face of column at section 1x-1x in T-m/m.

BM_{1z} =Moment at the face of column 1z-1z in T-m/m.

SF (One way)= Oneway shear force at a distance 'd' from the face of column in T/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in T/m.

Punching perimeter= ((0.55+2*0.5*0.201)+(0.55+2*0.5*0.201))*2

= 3.004 m

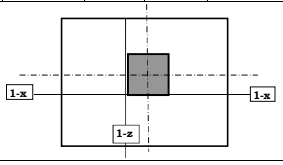
Punching Area= ((0.55+2*0.5*0.201)*(0.55+2*0.5*0.201))

= 0.56 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.3*2.5+1.2*1.8)

= 2.91 T/m^2

Area of footing, A_f= 3.24 m^2



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Date	05-06-2020	08-07-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 42.75 T

Strength of Concrete in shear Vc1 = 129.14 T — **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 139.21 T — **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 119.07 T — **Table 22.6.5.2_ (c)**

Design Vc= 119.07 T

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (42.75*10000)/(3004*201)

= 0.71 N/mm^2

No shear reinforcement reqd.

Load Case	p_{1x} (T/m^2)	B.M$_{1x}$ TM	S.F$_{(oneway)}$ (T)	B.M$_{1z}$	S.F$_{(Oneway)}$ (T)	S.F$_{(punching)}$	V_u/b_0d	Punching check (Remarks)	Allowable shear (0.75*$\sqrt{(fc)}$/6 (N/mm^2)	Actual Shear	Remarks	Mu/bd2
Ultimate Conditions												
1.4D	8.57	1.67	3.50	1.67	3.63	22.97	0.38	OK	0.74	0.181	OK	0.41
1.4D+1.7(CL1)+0.5Lr	12.86	2.51	5.25	2.51	5.45	34.46	0.57	OK	0.74	0.271	OK	0.62
1.4D+1.7(CL2)+0.5Lr	14.48	2.83	5.91	2.83	6.14	38.81	0.64	OK	0.74	0.305	OK	0.70
1.4D+1.7(CL3)+0.5Lr	12.08	2.36	4.93	2.36	5.12	32.37	0.54	OK	0.74	0.255	OK	0.58
1.4D+1.7(CL4)+0.5Lr	13.55	2.65	5.53	2.65	5.75	36.31	0.6	OK	0.74	0.286	OK	0.66
1.4D+1.7(CL5)+0.5Lr	14.57	2.85	5.94	2.85	6.18	39.05	0.65	OK	0.74	0.307	OK	0.70
1.4D+1.7(CL6)+0.5Lr	14.59	2.85	5.95	2.85	6.19	39.10	0.65	OK	0.74	0.308	OK	0.71
1.4D+1.7(CL5)+0.5Lr	13.71	2.68	5.59	2.68	5.81	36.74	0.61	OK	0.74	0.289	OK	0.66
1.4D+1.7(CL6)+0.5Lr	12.92	2.52	5.27	2.52	5.48	34.63	0.57	OK	0.74	0.273	OK	0.62
1.2D+1.6Lr+1.0(CL1)	14.93	2.92	6.09	2.92	6.33	40.01	0.66	OK	0.74	0.315	OK	0.72
1.2D+1.6Lr+1.0(CL2)	15.89	3.10	6.48	3.10	6.74	42.59	0.71	OK	0.74	0.335	OK	0.77
1.2D+1.6Lr+1.0(CL3)	14.47	2.83	5.90	2.83	6.14	38.78	0.64	OK	0.74	0.305	OK	0.70
1.2D+1.6Lr+1.0(CL4)	15.34	3.00	6.26	3.00	6.50	41.11	0.68	OK	0.74	0.324	OK	0.74
1.2D+1.6Lr+1.0(CL5)	15.94	3.11	6.50	3.11	6.76	42.72	0.71	OK	0.74	0.336	OK	0.77
1.2D+1.6Lr+1.0(CL6)	15.95	3.12	6.51	3.12	6.76	42.75	0.71	OK	0.74	0.336	OK	0.77
1.2D+1.6Lr+1.0(CL7)	15.43	3.01	6.30	3.01	6.54	41.35	0.68	OK	0.74	0.325	OK	0.75
1.2D+1.6Lr+1.0(CL8)	14.97	2.92	6.11	2.92	6.35	40.12	0.66	OK	0.74	0.316	OK	0.72
1.2D + 1.6WL1+1.0L+0.5Lr	6.86	1.34	2.80	1.34	2.91	18.38	0.3	OK	0.74	0.145	OK	0.33
1.2D + 1.6WL2+1.0L+0.5Lr	5.61	1.10	2.29	1.10	2.38	15.03	0.25	OK	0.74	0.118	OK	0.27
1.2D + 1.6WL3+1.0L+0.5Lr	10.44	2.04	4.26	2.04	4.43	27.98	0.46	OK	0.74	0.22	OK	0.50
1.2D + 1.6WL4+1.0L+0.5Lr	10.99	2.15	4.48	2.15	4.66	29.45	0.49	OK	0.74	0.232	OK	0.53
1.2D + 1.6WL5+1.0L+0.5Lr	12.16	2.38	4.96	2.38	5.16	32.59	0.54	OK	0.74	0.257	OK	0.59
1.2D + 1.6WL6+1.0L+0.5Lr	12.13	2.37	4.95	2.37	5.14	32.51	0.54	OK	0.74	0.256	OK	0.59
1.2D + 1.6WL7+1.0L+0.5Lr	12.71	2.48	5.19	2.48	5.39	34.06	0.56	OK	0.74	0.268	OK	0.61
1.2D + 1.6WL8+1.0L+0.5Lr	12.28	2.40	5.01	2.40	5.21	32.91	0.55	OK	0.74	0.259	OK	0.59
0.9D+1.6WL1	7.49	1.46	3.06	1.46	3.18	20.07	0.33	OK	0.74	0.158	OK	0.36
0.9D+1.6WL2	6.24	1.22	2.55	1.22	2.65	16.72	0.28	OK	0.74	0.132	OK	0.30
0.9D+1.6WL3	7.31	1.43	2.98	1.43	3.10	19.59	0.32	OK	0.74	0.154	OK	0.35
0.9D+1.6WL4	7.86	1.54	3.21	1.54	3.33	21.06	0.35	OK	0.74	0.166	OK	0.38
0.9D+1.6WL5	9.04	1.77	3.69	1.77	3.83	24.23	0.4	OK	0.74	0.191	OK	0.44
0.9D+1.6WL6	9.00	1.76	3.67	1.76	3.82	24.12	0.4	OK	0.74	0.19	OK	0.44
0.9D+1.6WL7	9.58	1.87	3.91	1.87	4.06	25.67	0.43	OK	0.74	0.202	OK	0.46
0.9D+1.6WL8	9.15	1.79	3.73	1.79	3.88	24.52	0.41	OK	0.74	0.193	OK	0.44

Load Case	kn (Ø=0.75)	p =	A_{st}
Ultimate Conditions			
1.4D	0.55	0.001	217
1.4D+1.7(CL1)+0.5Lr	0.83	0.002	434
1.4D+1.7(CL2)+0.5Lr	0.93	0.002	434
1.4D+1.7(CL3)+0.5Lr	0.78	0.002	434
1.4D+1.7(CL4)+0.5Lr	0.87	0.002	434
1.4D+1.7(CL5)+0.5Lr	0.94	0.002	434
1.4D+1.7(CL6)+0.5Lr	0.94	0.002	434
1.4D+1.7(CL5)+0.5Lr	0.88	0.002	434
1.4D+1.7(CL6)+0.5Lr	0.83	0.002	434
1.2D+1.6Lr+1.0(CL1)	0.96	0.002	434
1.2D+1.6Lr+1.0(CL2)	1.02	0.002	434
1.2D+1.6Lr+1.0(CL3)	0.93	0.002	434
1.2D+1.6Lr+1.0(CL4)	0.99	0.002	434
1.2D+1.6Lr+1.0(CL5)	1.03	0.002	434
1.2D+1.6Lr+1.0(CL6)	1.03	0.002	434
1.2D+1.6Lr+1.0(CL7)	0.99	0.002	434
1.2D+1.6Lr+1.0(CL8)	0.96	0.002	434
1.2D + 1.6WL1+1.0L+0.5Lr	0.44	0.001	217
1.2D + 1.6WL2+1.0L+0.5Lr	0.36	0.001	217
1.2D + 1.6WL3+1.0L+0.5Lr	0.67	0.002	434
1.2D + 1.6WL4+1.0L+0.5Lr	0.71	0.002	434
1.2D + 1.6WL5+1.0L+0.5Lr	0.78	0.002	434
1.2D + 1.6WL6+1.0L+0.5Lr	0.78	0.002	434
1.2D + 1.6WL7+1.0L+0.5Lr	0.82	0.002	434
1.2D + 1.6WL8+1.0L+0.5Lr	0.79	0.002	434
0.9D+1.6WL1	0.48	0.001	217
0.9D+1.6WL2	0.4	0.001	217
0.9D+1.6WL3	0.47	0.001	217
0.9D+1.6WL4	0.51	0.001	217
0.9D+1.6WL5	0.58	0.001	217
0.9D+1.6WL6	0.58	0.001	217
0.9D+1.6WL7	0.62	0.001	217
0.9D+1.6WL8	0.59	0.001	217

Location	Mux (T-m)	ϕ Kn = M_{u1}/bd^2	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing	Spacing provd.
At bottom	3.12	0.77	1.03	0.002	434.00	390.60	12ϕT	260.59	300	200
At top	0.00	0.00	0.00	0.0018	361.80	361.80	12ϕT	312.60	300	200

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction : At Bottom **12ϕT** **200** c/c At Top **12ϕT** **200** c/c

Along Z Direction : At Bottom **12ϕT** **200** c/c At Top **12ϕT** **200** c/c

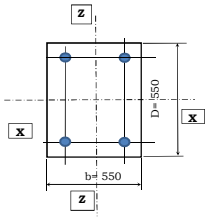
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	05-06-2020	08-07-2020	26-10-2023	Date	05-06-2020	08-07-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC	Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

DESIGN OF COLUMN MKD C1

Input:	
Clear cover to reinf.=	75mm
Dia. of main reinf.=	25mm
Dia. of link=	10mm
Effective cover =	97.5mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	0.50%
Max. Steel Ratio	4.00%
Pedestal Height	1700

Table 20.6.1.3.1



For Compression-controlled sections
Strength reduction factor ϕ= 0.65 Table 21.2.2

Unsuported length of column about X (l_{ux}) = 1700.00
Unsupported length of column about Z (l_{uz}) = 1700.00

For Columns not braced against sidesway,
Slenderness ratio, K l_{ux} / r_x = 21.42 < 22 SECTION 6.2.5a
K l_{uz} / r_z = 21.42 < 22 SECTION 6.2.5a

Where, Effective length factor K= 2
Radius of gyration r_x= 159
Radius of gyration r_z= 159

γ_r =	0.68
γ_s =	0.68

Load Combination	Factored Load/Moment (T)			Nominal Load/Moment			Kn = Pn/f_cbD (N/mm^2)
	Hx	V	Hz	Mz (Tm)	V (T)	Mx (Tm)	
1.4D	-1.58	9.44	0.00	4.1	14.5	0.0	0.01
1.4D+1.7(CL1)+0.5Lr	-2.45	17.50	0.00	6.4	26.9	0.0	0.03
1.4D+1.7(CL2)+0.5Lr	-3.15	18.10	0.00	8.2	27.8	0.0	0.03
1.4D+1.7(CL3)+0.5Lr	-2.49	14.75	0.00	6.5	22.7	0.0	0.02
1.4D+1.7(CL4)+0.5Lr	-3.12	15.33	0.00	8.2	23.6	0.0	0.02
1.4D+1.7(CL5)+0.5Lr	-2.80	17.79	0.44	7.3	27.4	1.2	0.03
1.4D+1.7(CL6)+0.5Lr	-2.81	17.81	-0.44	7.3	27.4	1.2	0.03
1.4D+1.7(CL5)+0.5Lr	-2.80	15.03	0.44	7.3	23.1	1.2	0.02
1.4D+1.7(CL6)+0.5Lr	-2.47	14.65	-0.44	6.5	22.5	1.2	0.02
1.2D+1.6Lr+1.0(CL1)	-3.37	18.12	0.00	8.8	27.9	0.0	0.03
1.2D+1.6Lr+1.0(CL2)	-3.78	18.47	0.00	9.9	28.4	0.0	0.03
1.2D+1.6Lr+1.0(CL3)	-3.39	16.50	0.00	8.9	25.4	0.0	0.02
1.2D+1.6Lr+1.0(CL4)	-3.76	16.84	0.00	9.8	25.9	0.0	0.02
1.2D+1.6Lr+1.0(CL5)	-3.57	18.29	0.26	9.3	28.1	0.7	0.03
1.2D+1.6Lr+1.0(CL6)	-3.58	18.30	-0.26	9.4	28.2	0.7	0.03
1.2D+1.6Lr+1.0(CL7)	-3.57	16.67	0.26	9.3	25.6	0.7	0.02
1.2D+1.6Lr+1.0(CL8)	-3.38	16.44	-0.26	8.8	25.3	0.7	0.02
1.2D + 1.6WL1+1.0L+0.5Lr	1.19	6.53	0.00	3.1	10.1	0.0	0.01
1.2D + 1.6WL2+1.0L+0.5Lr	0.82	4.94	0.00	2.1	7.6	0.0	0.01
1.2D + 1.6WL3+1.0L+0.5Lr	-2.54	9.08	0.00	6.6	14.0	0.0	0.01
1.2D + 1.6WL4+1.0L+0.5Lr	-3.00	7.85	0.00	7.8	12.1	0.0	0.01
1.2D + 1.6WL5+1.0L+0.5Lr	-1.78	7.25	1.87	4.7	11.2	4.9	0.01
1.2D + 1.6WL6+1.0L+0.5Lr	-2.06	5.29	-1.87	5.4	8.1	4.9	0.01
1.2D + 1.6WL7+1.0L+0.5Lr	-2.00	7.55	1.87	5.2	11.6	4.9	0.01
1.2D + 1.6WL8+1.0L+0.5Lr	-2.12	5.41	-1.87	5.5	8.3	4.9	0.01
0.9D+1.6WL1	2.10	2.49	0.00	5.5	3.8	0.0	0.00
0.9D+1.6WL2	1.73	0.90	0.00	4.5	1.4	0.0	0.00
0.9D+1.6WL3	-1.63	5.04	0.00	4.3	7.7	0.0	0.01
0.9D+1.6WL4	-2.08	3.80	0.00	5.4	5.8	0.0	0.01
0.9D+1.6WL5	-0.87	3.21	1.87	2.3	4.9	4.9	0.00
0.9D+1.6WL6	-1.15	1.24	-1.87	3.0	1.9	4.9	0.00
0.9D+1.6WL7	-1.09	3.51	1.87	2.8	5.4	4.9	0.01
0.9D+1.6WL8	-1.20	1.37	-1.87	3.1	2.1	4.9	0.00

REINFORCEMENT

Provide,

4-Nos	25 mm Dia	1964.29 Sq mm	(At Corner)	A_{sc} pvd. =	3221.43 Sq mm
2-Nos	20 mm Dia	628.57 Sq mm	(At middle)		
2-Nos	20 mm Dia	628.57 Sq mm	(At middle)	Total A_{sc} =	1.0 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc} = 1025.65375 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.7		M_{noz}	M_{nox}	β	α_n = log(0.5)/ log(β)	Interaction ratio	Remarks
	R_{nz}=M_{noz} /$f'_c b^2 D$	R_{nx}=M_{nox} /$f'_c bD^2$						
1.4D	0.06	0.06	34.9	34.94	0.55	1.16	0.08	O.K
1.4D+1.7(CL1)+0.5Lr	0.07	0.07	40.8	40.76	0.55	1.16	0.12	O.K
1.4D+1.7(CL2)+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.19	O.K
1.4D+1.7(CL3)+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.14	O.K
1.4D+1.7(CL4)+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.19	O.K
1.4D+1.7(CL5)+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.18	O.K
1.4D+1.7(CL6)+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.18	O.K
1.4D+1.7(CL5)+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.18	O.K
1.4D+1.7(CL6)+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.16	O.K
1.2D+1.6Lr+1.0(CL1)	0.06	0.06	34.9	34.94	0.55	1.16	0.20	O.K
1.2D+1.6Lr+1.0(CL2)	0.06	0.06	34.9	34.94	0.55	1.16	0.23	O.K
1.2D+1.6Lr+1.0(CL3)	0.06	0.06	34.9	34.94	0.55	1.16	0.20	O.K
1.2D+1.6Lr+1.0(CL4)	0.06	0.06	34.9	34.94	0.55	1.16	0.23	O.K
1.2D+1.6Lr+1.0(CL5)	0.06	0.06	34.9	34.94	0.55	1.16	0.23	O.K
1.2D+1.6Lr+1.0(CL6)	0.06	0.06	34.9	34.94	0.55	1.16	0.23	O.K
1.2D+1.6Lr+1.0(CL5)	0.06	0.06	34.9	34.94	0.55	1.16	0.23	O.K
1.2D+1.6Lr+1.0(CL6)	0.06	0.06	34.9	34.94	0.55	1.16	0.21	O.K
1.2D + 1.6WL1+1.0L+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.06	O.K
1.2D + 1.6WL2+1.0L+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.04	O.K
1.2D + 1.6WL3+1.0L+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.15	O.K
1.2D + 1.6WL4+1.0L+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.18	O.K
1.2D + 1.6WL5+1.0L+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.20	O.K
1.2D + 1.6WL6+1.0L+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.22	O.K
1.2D + 1.6WL7+1.0L+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.21	O.K
1.2D + 1.6WL8+1.0L+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.22	O.K
0.9D+1.6WL1	0.06	0.06	34.9	34.94	0.55	1.16	0.12	O.K
0.9D+1.6WL2	0.06	0.06	34.9	34.94	0.55	1.16	0.09	O.K
0.9D+1.6WL3	0.06	0.06	34.9	34.94	0.55	1.16	0.09	O.K
0.9D+1.6WL4	0.06	0.06	34.9	34.94	0.55	1.16	0.12	O.K
0.9D+1.6WL5	0.05	0.05	29.1	29.12	0.55	1.16	0.18	O.K
0.9D+1.6WL6	0.05	0.05	29.1	29.12	0.55	1.16	0.20	O.K
0.9D+1.6WL7	0.05	0.05	29.1	29.12	0.55	1.16	0.19	O.K
0.9D+1.6WL8	0.05	0.05	29.1	29.12	0.55	1.16	0.20	O.K

Links, **10mm dia @ 300 C/C**

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	05-06-2020	08-07-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	05-06-2020	08-07-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

2.0 DESIGN OF FOUNDATION MKD F2

	C-1			C-2		
LOAD ITEM	Fx (T)	Fy (T)	Fz (T)	Fx (T)	Fy (T)	Fz (T)
1 DL	0.97	7.17	0.04	1.05	6.68	-0.04
2 LLROOF	0.73	2.37	0.00	1.13	4.01	0.00
3 CL1(L+XS)	0.57	2.29	0.00	0.57	2.28	0.00
4 CL2(L-XS)	0.20	1.94	0.00	0.20	1.94	0.00
5 CL3(R+XS)	0.59	3.91	0.00	0.59	3.91	0.00
6 CL4(R-XS)	0.18	3.57	0.00	0.18	3.56	0.00
7 CL5(L+ZT)	0.39	2.56	0.13	0.38	1.66	0.13
8 CL6(L-ZT)	0.38	1.66	-0.13	0.38	2.56	-0.13
9 CL7(R+ZT)	0.39	4.19	0.13	0.38	3.29	0.13
10 CL8(R-ZT)	0.10	-0.41	-0.16	0.10	0.67	-0.16
21 WL1(+X)S	0.68	-0.60	0.18	0.43	-0.64	0.09
22 WL2(+X)P	0.69	-1.75	0.42	0.66	-1.76	0.19
23 WL3(-X)S	-1.91	-2.21	0.17	-1.91	-2.21	0.08
24 WL4(-X)P	-1.71	-3.26	0.41	-1.71	-3.23	0.19
25 WL5(+Z)S	-0.06	1.11	0.92	-0.08	-4.57	0.87
26 WL6(+Z)P	0.11	-0.50	1.05	0.08	-5.55	0.88
27 WL7(-Z)S	-0.17	-4.78	-1.26	-0.16	1.09	-1.03
28 WL8(-Z)P	0.06	-5.60	-0.89	0.09	-0.46	-0.81

FOUNDATION SIZE	2.00m	X	4.25m	
DEPTH OF THE FOUNDATION	1.50m	(below FGL)		
THE WT OF SOIL +SELF WT	2X4.25X1.5XUNIT WT		=	25.5 T

WORKING STRESS DESIGN

	C-1			C-2			Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
DL+Lr	1.36	10.04	0.05	1.47	9.35	-0.05	44.89	2.83	0.93	0.00	5.66
DL+CL1	2.69	15.11	0.06	3.01	15.24	-0.05	55.84	5.70	0.18	0.00	11.40
DL+CL2	2.06	14.52	0.06	2.37	14.65	-0.05	54.67	4.44	0.18	0.00	8.87
DL+CL3	2.73	17.87	0.06	3.04	18.00	-0.05	61.37	5.76	0.18	0.00	11.53
DL+CL4	2.03	17.28	0.06	2.34	17.41	-0.05	60.20	4.38	0.17	0.00	8.75
DL+CL5	2.38	15.58	0.28	2.69	14.18	0.17	55.26	5.07	2.78	0.44	10.14
DL+CL6	2.37	14.05	-0.17	2.69	15.71	-0.28	55.26	5.07	3.13	-0.44	10.13
DL+CL5	2.38	18.35	0.28	2.69	16.94	0.17	60.79	5.07	2.78	0.44	10.14
DL+CL6	1.90	10.53	-0.21	2.21	12.50	-0.32	48.53	4.10	3.73	-0.53	8.21
DL+0.75*CL1+0.75*Lr	2.90	14.68	0.05	3.64	16.72	-0.05	56.89	6.54	2.76	0.00	13.08
DL+0.75*CL2+0.75*Lr	2.53	14.33	0.05	3.27	16.37	-0.05	56.20	5.80	2.76	0.00	11.59
DL+0.75*CL3+0.75*Lr	2.92	16.30	0.05	3.66	18.34	-0.05	60.14	6.58	2.76	0.00	13.15
DL+0.75*CL4+0.75*Lr	2.51	15.96	0.05	3.25	18.00	-0.05	59.45	5.76	2.76	0.00	11.52
DL+0.75*CL5+0.75*Lr	2.71	14.96	0.18	3.46	16.10	0.08	56.55	6.17	2.06	0.26	12.34
DL+0.75*CL6+0.75*Lr	2.71	14.05	-0.08	3.46	17.00	-0.18	56.55	6.17	4.49	-0.26	12.33
DL+0.75*CL5+0.75*Lr	2.71	16.58	0.18	3.46	17.72	0.08	59.80	6.17	2.06	0.26	12.34
DL+0.75*CL6+0.75*Lr	2.43	11.98	-0.11	3.17	15.11	-0.20	52.59	5.60	4.84	-0.31	11.20
DL+0.75*Lr+0.75*WL1	2.62	8.83	0.33	2.51	8.99	0.10	43.32	5.13	1.08	0.43	10.27
DL+0.75*Lr+0.75*WL2	2.63	6.99	0.71	2.88	7.21	0.26	39.70	5.51	2.25	0.97	11.01
DL+0.75*Lr+0.75*WL3	-1.53	6.25	0.31	-1.23	6.48	0.08	38.23	-2.75	1.09	0.39	5.51
DL+0.75*Lr+0.75*WL4	-1.20	4.58	0.70	-0.90	4.85	0.25	34.93	-2.11	2.28	0.96	4.21
DL+0.75*Lr+0.75*WL5	1.44	11.6	1.5	1.71	2.71	1.34	39.77	3.14	17.68	2.86	6.28
DL+0.75*Lr+0.75*WL6	1.71	9.0	1.7	1.96	1.14	1.36	35.63	3.67	16.76	3.08	7.34
DL+0.75*Lr+0.75*WL7	1.25	2.1	-2.0	1.57	11.76	-1.70	39.40	2.82	20.32	-3.67	5.64
DL+0.75*Lr+0.75*WL8	1.63	0.8	-1.4	1.97	9.28	-1.33	35.61	3.59	16.85	-2.72	7.19
0.6*DL+WL1	1.97	5.5	0.3	1.63	4.98	0.11	35.97	3.60	1.55	0.43	7.20
0.6*DL+WL2	1.97	3.7	0.7	2.00	3.20	0.27	32.35	3.97	2.55	0.97	7.94
0.6*DL+WL3	-2.18	2.9	0.3	-2.11	2.47	0.09	30.89	-4.29	1.37	0.39	8.58
0.6*DL+WL4	-1.86	1.2	0.7	-1.79	0.84	0.26	27.59	-3.64	2.45	0.95	7.29
0.6*DL+WL5	0.78	8.2	1.5	0.82	-1.30	1.35	32.42	1.61	18.59	2.86	3.21
0.6*DL+WL6	1.06	5.7	1.7	1.08	-2.87	1.37	28.29	2.14	17.67	3.08	4.27
0.6*DL+WL7	0.60	-1.2	-2.0	0.69	7.75	-1.69	32.06	1.29	19.41	-3.67	2.57
0.6*DL+WL8	0.97	-2.5	-1.4	1.09	5.27	-1.32	28.26	2.06	15.94	-2.72	4.12

Check for overturning and sliding:

	Load at Foundation Base					Sliding Coeffecient, μ = 0.4			
Load Case	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
DL+Lr	44.89	2.83	5.66	0.00	0.93	2181.08	102.99	6.34	-
DL+CL1	55.84	5.70	11.40	0.00	0.18	1346.77	670.55	3.92	34365.99
DL+CL2	54.67	4.44	8.87	0.00	0.18	1694.14	656.48	4.93	33645.19
DL+CL3	61.37	5.76	11.53	0.00	0.18	1464.17	724.69	4.26	37767.35
DL+CL4	60.20	4.38	8.75	0.00	0.17	1891.65	735.19	5.50	37043.83
DL+CL5	55.26	5.07	10.14	0.44	2.78	1498.92	42.29	4.36	49.94
DL+CL6	55.26	5.07	10.13	-0.44	3.13	1500.05	37.53	4.36	50.08
DL+CL5	60.79	5.07	10.14	0.44	2.78	1648.83	46.46	4.80	54.93
DL+CL6	48.53	4.10	8.21	-0.53	3.73	1625.90	27.63	4.73	36.34
DL+0.75*CL1+0.75*Lr	56.89	6.54	13.08	0.00	2.76	1196.00	43.81	3.48	10941.00
DL+0.75*CL2+0.75*Lr	56.20	5.80	11.59	0.00	2.76	1333.07	43.28	3.88	10808.50
DL+0.75*CL3+0.75*Lr	60.14	6.58	13.15	0.00	2.76	1257.35	46.28	3.66	11566.25
DL+0.75*CL4+0.75*Lr	59.45	5.76	11.52	0.00	2.76	1419.03	45.81	4.13	11433.25
DL+0.75*CL5+0.75*Lr	56.55	6.17	12.34	0.26	2.06	1260.46	58.23	3.67	86.31
DL+0.75*CL6+0.75*Lr	56.55	6.17	12.33	-0.26	4.49	1260.90	26.77	3.67	87.70
DL+0.75*CL5+0.75*Lr	59.80	6.17	12.34	0.26	2.06	1332.93	61.63	3.88	91.27
DL+0.75*CL6+0.75*Lr	52.59	5.60	11.20	-0.31	4.84	1291.00	23.07	3.76	67.31
DL+0.75*Lr+0.75*WL1	43.32	5.13	10.27	0.43	1.08	1160.37	85.25	3.38	40.38
DL+0.75*Lr+0.75*WL2	39.70	5.51	11.01	0.97	2.25	991.50	37.43	2.88	16.30
DL+0.75*Lr+0.75*WL3	38.23	-2.75	5.51	0.39	1.09	1908.74	74.79	5.55	39.46
DL+0.75*Lr+0.75*WL4	34.93	-2.11	4.21	0.96	2.28	2279.12	32.50	6.63	14.63
DL+0.75*Lr+0.75*WL5	39.77	3.14	6.28	2.86	17.68	1739.94	4.78	5.06	5.56
DL+0.75*Lr+0.75*WL6	35.63	3.67	7.34	3.08	16.76	1334.63	4.52	3.88	4.62
DL+0.75*Lr+0.75*WL7	39.40	2.82	5.64	-3.67	20.32	1919.69	4.12	5.58	4.29
DL+0.75*Lr+0.75*WL8	35.61	3.59	7.19	-2.72	16.85	1362.34	4.49	3.96	5.24
0.6*DL+WL1	35.97	3.60	7.20	0.43	1.55	1374.86	49.39	4.00	33.58
0.6*DL+WL2	32.35	3.97	7.94	0.97	2.55	1120.57	26.92	3.26	13.29
0.6*DL+WL3	30.89	-4.29	8.58	0.39	1.37	990.16	47.74	2.88	31.94
0.6*DL+WL4	27.59	-3.64	7.29	0.95	2.45	1041.32	23.94	3.03	11.56
0.6*DL+WL5	32.42	1.61	3.21	2.86	18.59	2773.72	3.71	8.07	4.53
0.6*DL+WL6	28.29	2.14	4.27	3.08	17.67	1821.26	3.40	5.30	3.67
0.6*DL+WL7	32.06	1.29	2.57	-3.67	19.41	3425.25	3.51	9.96	3.49
0.6*DL+WL8	28.26	2.06	4.12	-2.72	15.94	1887.83	3.77	5.49	4.16

CHECK FOR BEARING PRESSURE

THE NET SAFE BEARING CAPACITY OF SOIL IS 50T/m^2

MAXIMUM GROSS BEARING PRESSURE OBTAINED 18T/m^2 IN LC 3 . HENCE IT IS SAFE.

REFER TO THE STAAD FILE ATTACHED.

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	05-06-2020	08-07-2020	26-10-2023	Date	05-06-2020	08-07-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC	Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

Foundation has been designed with the Design strength Load Combinations of Staad File (LC 100 - LC 132) (REFER STAAD MODEL)

REINFORCEMENT CALCULATION OF FOUNDATION:

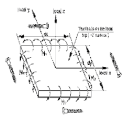
Foundation Thickness =	300	mm
Width =	1000	mm
Clear Cover =	75	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	1.201	T-M	Corresponding Mxy	-0.87	T-M	*(Plate No. - 31 , L/C No. - 111)*
Min. Mx	-2.471	T-M	Corresponding Mxy	-0.64	T-M	*(Plate No. - 82 , L/C No. - 111)*
Max. My	2.62	T-M	Corresponding Mxy	-0.20	T-M	*(Plate No. - 108 , L/C No. - 111)*
Min. My	-2.055	T-M	Corresponding Mxy	-0.64	T-M	*(Plate No. - 82 , L/C No. - 111)*

FOR OPPOSITE FACE REINFORCEMENT :

Mx1 =	Mx + abs(Mxy) =	1.201 + 0.87 =	2.071	T-M
My1 =	My + abs(Mxy) =	2.62 + 0.203 =	2.823	T-M
Mx2 =	Mx+abs(Mxy^2/My) =	1.201 + 0.289 =	1.49	T-M
My2 =	My+abs(Mxy^2/Mx) =	2.62 + 0.034 =	2.654	T-M



DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT (Bottom) :

Mxd = 2.071 T-m
Myd = 2.823 T-m

ALONG X - DIRECTION :

Mu = 2.071 T-m
Φ*Kn = Mu/bd^2 deff. = 207 mm
Mu/bd^2 = 2.071X10^7/(1000X207^2) = 0.483 N/mm^2 < 6.9 N/mm^2
Pt = 0.0013 Reqd. Ast = 0.0013X1000X207/100 = 2.69 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 2.823 T-m
Φ*Kn = Mu/bd^2 deff. = 219 mm
Mu/bd^2 = 2.823X10^7/(1000X219^2) = 0.589 N/mm^2 < 6.9 N/mm^2
Pt = 0.0016 Reqd. Ast = 0.0016X1000X219/100 = 3.5 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

FOR RETAINED FACE REINFORCEMENT :

Mx1 =	Mx - abs(Mxy) =	-2.471 - 0.643 =	-3.114	T-M
My1 =	My - abs(Mxy) =	-2.055 - 0.643 =	-2.698	T-M
Mx2 =	Mx-abs(Mxy^2/My) =	-2.471 - 0.201 =	-2.672	T-M
My2 =	My-abs(Mxy^2/Mx) =	-2.055 - 0.167 =	-1.888	T-M

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT (Top) :

Mxd = -3.114 T-M
Myd = -2.698 T-M

ALONG X - DIRECTION :

Mu = 3.114 T-m
Φ*Kn = Mu/bd^2 deff. = 207 mm
Mu/bd^2 = 3.114X10^7/(1000X207^2) = 0.727 N/mm^2 < 6.9 N/mm^2
Pt = 0.0020 Reqd. Ast = 0.002X1000X207/100 = 4.14 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 2.698 T-m
Φ*Kn = Mu/bd^2 deff. = 219 mm
Mu/bd^2 = 2.698X10^7/(1000X219^2) = 0.563 N/mm^2 < 6.9 N/mm^2
Pt = 0.0015 Reqd. Ast = 0.0015X1000X219/100 = 3.29 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	05-06-2020	08-07-2020	26-10-2023				Date	05-06-2020	08-07-2020	26-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

SHEAR CHECK

(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

SQx = 0.172 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 83 , L/C No. - 111)*

Vu = 3.56 T

Provided Pt = 0.0027

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0027X3.56X10^4X207/3.114/10^7)X1000X207/10^4

= 19.819 T

[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0027)X1000X207/10^4

= 20.544 T

[c] = 0.29*λ*fc^0.5*b*d.eff

= 0.29X1X5.92X1000X207/10^4

= 35.514 T

HENCE, Vc = 19.819 T

Φt*Vc = 0.75X19.819 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 14.864 T **> 3.56T , Hence O.K.**

ALONG Y - DIRECTION :

SQy = 0.163 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 93 , L/C No. - 111)*

Vu = 3.57 T

Pt provided = 0.0026

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0026X3.57X10^4X219/2.698/10^7)X1000X219/10^4

= 21.01 T

[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0026)X1000X219/10^4

= 21.698 KN

[c] = 0.29*λ*fc^0.5*b*d.eff

= 0.29X1X5.92X1000X219/10^4

= 37.573 T

HENCE, Vc = 21.01 T

Φt*Vc = 0.75X21.01 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 15.758 T **> 3.57T , Hence O.K.**

PEDESTAL REINFORCEMENT

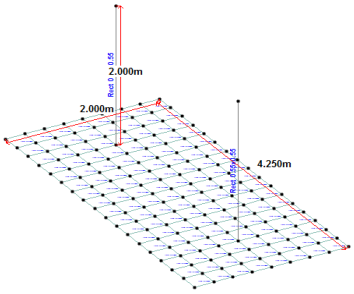
Size of Column = 550 X 550 mm^2

Reqired area of Reinforcement = 3025 mm^2

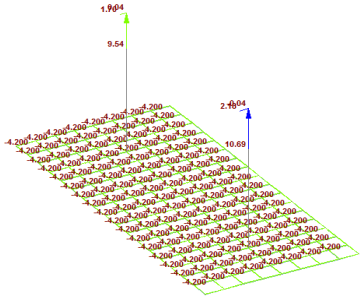
Reinforcement Provided = 4 no. 20 dia bars + 4 no. 25 dia bars

Area of Reinforcement Provided = 3220.1 mm^2

STAAD MODEL OF RAFT WITH PLATE ELEMENT



STAAD MODEL OF RAFT WITH DEAD + LIVE+ SURCHARGE LOAD



Surcharge Load on Foundation = 2 T/m^2

Soil Weight over the foundation = 1.2x 1.8 T/m2 = 2.16 T/m^2

SOIL+SURCHARGE LOAD = 4.16 T/m^2

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01
Date	05-06-2020	08-07-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	05-06-2020	08-07-2020	26-10-2023
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Approved	PRI	PRI	PRI

REFERENCES

[1] Structural Loading and Forces ; SBC Code 301: 2007
[2] Saudi Concrete Structures Code ; SBC 304 : 2018
[3] Civil,Structural and Architectural Design Criteria ; MD-216-1000-EG-ST-CRT-1001
[4] ACI 318 : 2014 Building Code Requirement of Structural Concrete
[5] Geotechnical Report No. MD-216-3000-EG-CV-RPT-1008

Global Project ID	**900707**
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Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
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Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	05-06-2020	08-07-2020	26-10-2023				Date	05-06-2020	08-07-2020	26-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 1 Foundation Load Data

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
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Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
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Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC	Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

MMGP:POWER PLANT:FUEL TREATMENT HOUSE LOAD DATA (DATED 01.06.2020)

1 DL	Dead load		
2 LLROOF	Live load on roof		
3 CL1(L+XS)	Crane load	Max Column concentration at B grid,Surge in direction in +X (from B grid to F grid)	
4 CL2(L-XS)	Crane load	Max Column concentration at B grid,Surge in direction in -X (from F grid to B grid)	
5 CL3(R+XS)	Crane load	Max Column concentration at F grid,Surge in direction in +X (from B grid to F grid)	
6 CL4(R-XS)	Crane load	Max Column concentration at F grid,Surge in direction in -X (from F grid to B grid)	
7 CL5(L+ZT)	Crane load	Max Column concentration at B grid,tractive in direction in +Z (from line-1 to line-9)	
8 CL6(L-ZT)	Crane load	Max Column concentration at B grid,tractive in direction in -Z (from line-9 to line-1)	
9 CL7(R+ZT)	Crane load	Max Column concentration at F grid,tractive in direction in +Z (from line-1 to line-9)	
10 CL8(R-ZT)	Crane load	Max Column concentration at F grid,tractive in direction in -Z (from line-9 to line-1)	
21 WL1(+X)S	Wind load	Wind load in direction of +X Suction(from A grid to F grid)	side wind
22 WL2(+X)P	Wind load	Wind load in direction of +X Pressure(from A grid to F grid)	side wind
23 WL3(-X)S	Wind load	Wind load in direction of -X Suction(from F grid to A grid)	side wind
24 WL4(-X)P	Wind load	Wind load in direction of -X Pressure(from F grid to A grid)	side wind
25 WL5(+Z)S	Wind load	Wind load in direction of +Z Suction(from line-1 to line-6)	gable wind
26 WL6(+Z)P	Wind load	Wind load in direction of +Z Pressure(from line-1 to line-6)	gable wind
27 WL7(-Z)S	Wind load	Wind load in direction of -Z Suction(from line-6 to line-1)	gable wind
28 WL8(-Z)P	Wind load	Wind load in direction of -Z Pressure(from line-6 to line-1)	gable wind

All crane and wind load cases are exclusive.

Sign convention

FY	+ve sign indicates vertical load downward
	-ve sign indicates vertical load upward
FX	+ve sign indicates horizonal load in across direction from grid-A to F
	-ve sign indicates horizonal load in across direction from grid-F to A
FZ	+ve sign indicates horizonal load in along direction from line 1 to 9
	-ve sign indicates horizonal load in along direction from line 9 to 1

LOAD DATA TABLE

	Load cases	Horizontal Fx Ton	Vertical Fy Ton	Horizontal Fz Ton	
A	1 DL	-1.1	6.7	0.0	MAIN COLUMN(3,4,5)
	2 LLROOF	-1.1	4.0	0.0	
	3 CL1(L+XS)	-0.2	3.6	0.0	
	4 CL2(L-XS)	-0.6	3.9	0.0	
	5 CL3(R+XS)	-0.2	1.9	0.0	
	6 CL4(R-XS)	-0.6	2.3	0.0	
	7 CL5(L+ZT)	-0.4	3.7	0.3	
	8 CL6(L-ZT)	-0.4	3.7	-0.3	
	9 CL7(R+ZT)	-0.4	2.1	0.3	
	10 CL8(R-ZT)	-0.2	1.9	-0.3	
	21 WL1(+X)S	1.9	-2.2	0.0	
	22 WL2(+X)P	1.7	-3.2	0.0	
	23 WL3(-X)S	-0.4	-0.6	0.0	
	24 WL4(-X)P	-0.7	-1.4	0.0	
	25 WL5(+Z)S	0.1	-1.8	1.2	
	26 WL6(+Z)P	-0.1	-3.0	-1.2	
	27 WL7(-Z)S	0.0	-1.6	1.2	
	28 WL8(-Z)P	-0.1	-2.9	-1.2	
C	1 DL	1.1	6.7	0.0	
	2 LLROOF	1.1	4.0	0.0	
	3 CL1(L+XS)	0.6	2.3	0.0	
	4 CL2(L-XS)	0.2	1.9	0.0	
	5 CL3(R+XS)	0.6	3.9	0.0	
	6 CL4(R-XS)	0.2	3.6	0.0	
	7 CL5(L+ZT)	0.4	2.1	0.3	
	8 CL6(L-ZT)	0.4	2.1	-0.3	
	9 CL7(R+ZT)	0.4	3.7	0.3	
	10 CL8(R-ZT)	0.1	0.2	-0.3	
	21 WL1(+X)S	0.4	-0.6	0.0	
	22 WL2(+X)P	0.7	-1.7	0.0	
	23 WL3(-X)S	-1.9	-2.2	0.0	
	24 WL4(-X)P	-1.7	-3.2	0.0	
	25 WL5(+Z)S	-0.1	-1.8	1.2	
	26 WL6(+Z)P	0.1	-3.0	-1.2	
	27 WL7(-Z)S	-0.1	-1.8	1.2	
	28 WL8(-Z)P	0.1	-2.9	-1.2	
gable post (B1,B5)	1 DL	0.0	2.3	0.0	Gable post
	2 LLROOF	0.0	0.0	0.0	
	3 CL1(L+XS)	0.0	0.0	0.0	
	4 CL2(L-XS)	0.0	0.0	0.0	
	5 CL3(R+XS)	0.0	0.0	0.0	
	6 CL4(R-XS)	0.0	0.0	0.0	
	7 CL5(L+ZT)	0.0	0.0	0.0	
	8 CL6(L-ZT)	0.0	0.0	0.0	
	9 CL7(R+ZT)	0.0	0.0	0.0	
	10 CL8(R-ZT)	0.0	0.0	0.0	
	21 WL1(+X)S	0.0	0.0	0.6	
	22 WL2(+X)P	0.0	0.0	1.4	
	23 WL3(-X)S	0.0	0.0	0.6	
	24 WL4(-X)P	0.0	0.0	1.4	
	25 WL5(+Z)S	0.0	0.0	0.4	
	26 WL6(+Z)P	0.0	0.0	1.1	
	27 WL7(-Z)S	0.0	0.0	-1.4	
	28 WL8(-Z)P	0.0	0.0	-0.5	

For layout drg and base plate and cross ection
Ref. MD-216-49PR-EG-ST-GAD-2405

Global Project ID	900707
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Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

A	1 DL	-1.1	6.7	0.0	MAIN COLUMN(1)	
	2 LLROOF	-1.1	4.0	0.0		
	3 CL1(L+XS)	-0.2	3.6	0.0		
	4 CL2(L-XS)	-0.6	3.9	0.0		
	5 CL3(R+XS)	-0.2	1.9	0.0		
	6 CL4(R-XS)	-0.6	2.3	0.0		
	7 CL5(L+ZT)	-0.4	3.3	0.1		
	8 CL6(L-ZT)	-0.4	4.2	-0.1		
	9 CL7(R+ZT)	-0.4	1.7	0.1		
	10 CL8(R-ZT)	-0.1	1.8	0.0		
	21 WL1(+X)S	2.0	-2.2	0.1		
	22 WL2(+X)P	1.7	-3.3	0.2		
	23 WL3(-X)S	-0.4	-0.7	0.1		
	24 WL4(-X)P	-0.6	-1.5	0.2		
	25 WL5(+Z)S	0.1	-4.6	0.9		
	26 WL6(+Z)P	-0.1	-5.5	0.9		
	27 WL7(-Z)S	0.0	1.1	-1.0		
	28 WL8(-Z)P	-0.1	-0.5	-0.8		
C	1 DL	1.1	6.7	0.0		
	2 LLROOF	1.1	4.0	0.0		
	3 CL1(L+XS)	0.6	2.3	0.0		
	4 CL2(L-XS)	0.2	1.9	0.0		
	5 CL3(R+XS)	0.6	3.9	0.0		
	6 CL4(R-XS)	0.2	3.6	0.0		
	7 CL5(L+ZT)	0.4	1.7	0.1		
	8 CL6(L-ZT)	0.4	2.6	-0.1		
	9 CL7(R+ZT)	0.4	3.3	0.1		
	10 CL8(R-ZT)	0.1	0.7	-0.2		
	21 WL1(+X)S	0.4	-0.6	0.1		
	22 WL2(+X)P	0.7	-1.8	0.2		
	23 WL3(-X)S	-1.9	-2.2	0.1		
	24 WL4(-X)P	-1.7	-3.2	0.2		
	25 WL5(+Z)S	-0.1	-4.6	0.9		
	26 WL6(+Z)P	0.1	-5.5	0.9		
	27 WL7(-Z)S	-0.2	1.1	-1.0		
	28 WL8(-Z)P	0.1	-0.5	-0.8		
A	1 DL	-1.0	7.2	0.0	MAIN COLUMN(2)	
	2 LLROOF	-0.7	2.4	0.0		
	3 CL1(L+XS)	-0.2	3.6	0.0		
	4 CL2(L-XS)	-0.6	3.9	0.0		
	5 CL3(R+XS)	-0.2	1.9	0.0		
	6 CL4(R-XS)	-0.6	2.3	0.0		
	7 CL5(L+ZT)	-0.4	4.2	0.1		
	8 CL6(L-ZT)	-0.4	3.3	-0.1		
	9 CL7(R+ZT)	-0.4	2.6	0.1		
	10 CL8(R-ZT)	-0.1	1.8	0.0		
	21 WL1(+X)S	1.9	-2.3	0.2		
	22 WL2(+X)P	1.7	-3.3	0.4		
	23 WL3(-X)S	-0.4	-0.7	0.2		
	24 WL4(-X)P	-0.6	-1.8	0.4		
	25 WL5(+Z)S	0.1	1.1	0.9		
	26 WL6(+Z)P	-0.1	-0.5	1.0		
	27 WL7(-Z)S	-0.4	-4.5	-1.2		
	28 WL8(-Z)P	-0.1	-5.6	-0.9		
C	1 DL	1.0	7.2	0.0		
	2 LLROOF	0.7	2.4	0.0		
	3 CL1(L+XS)	0.6	2.3	0.0		
	4 CL2(L-XS)	0.2	1.9	0.0		
	5 CL3(R+XS)	0.6	3.9	0.0		
	6 CL4(R-XS)	0.2	3.6	0.0		
	7 CL5(L+ZT)	0.4	2.6	0.1		
	8 CL6(L-ZT)	0.4	1.7	-0.1		
	9 CL7(R+ZT)	0.4	4.2	0.1		
	10 CL8(R-ZT)	0.1	-0.4	-0.2		
	21 WL1(+X)S	0.7	-0.6	0.2		
	22 WL2(+X)P	0.7	-1.8	0.4		
	23 WL3(-X)S	-1.9	-2.2	0.2		
	24 WL4(-X)P	-1.7	-3.3	0.4		
	25 WL5(+Z)S	-0.1	1.1	0.9		
	26 WL6(+Z)P	0.1	-0.5	1.0		
	27 WL7(-Z)S	-0.2	-4.8	-1.3		
	28 WL8(-Z)P	0.1	-5.6	-0.9		

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Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
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Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 2 STAAD Input & Output File of Foundation Mkd. F2

```
**************************************************
*                                                *
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*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    JUN  5, 2020                *
*           Time=    11:23:52                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
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INPUT FILE: C1-C2_FDN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 02-APR-19
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
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48. ELEMENT INCIDENCES SHELL
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72. 124 131 132 136 135; 125 76 133 137 81; 126 133 134 138 137
73. 127 134 135 139 138; 128 135 136 140 139; 129 81 137 141 86
74. 130 137 138 142 141; 131 138 139 143 142; 132 139 140 144 143
75. 135 83 82 147 148; 136 82 84 149 147; 137 84 85 150 149; 138 85 86 151 150
76. 139 86 141 152 151; 140 141 142 153 152; 141 142 143 154 153
77. 142 143 144 155 154; 143 156 157 21 20; 144 157 158 24 21; 145 158 159 26 24
78. 146 159 160 16 26; 147 160 161 90 16; 148 161 162 92 90; 149 162 163 94 92
79. 150 163 164 18 94; 151 165 166 157 156; 152 166 167 158 157
80. 153 167 168 159 158; 154 168 169 160 159; 155 169 170 161 160
81. 156 170 171 162 161; 157 171 172 163 162; 158 172 173 164 163
82. 159 148 147 174 175; 160 147 149 176 174; 161 149 150 177 176
83. 162 150 151 178 177; 163 151 152 179 178; 164 152 153 180 179
84. 165 153 154 181 180; 166 154 155 182 181
85. START GROUP DEFINITION
86. ELEMENT
87. _MOMENT 25 TO 27 29 TO 31 33 TO 71 73 TO 75 82 TO 84 86 TO 124 126 TO 128 -
88. 130 TO 132 135 TO 166
89. _SHEAR 25 26 29 30 33 34 37 TO 66 69 70 73 74 83 84 87 88 91 TO 120 123 124 -
90. 127 128 131 132 135 136 141 TO 144 149 TO 166
91. END GROUP DEFINITION
92. ELEMENT PROPERTY
93. 25 TO 76 81 TO 132 135 TO 166 THICKNESS 0.3
94. SUPPORTS
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95. 13 14 16 18 20 TO 86 90 TO 144 147 TO 181 -
96. 182 ELASTIC MAT DIRECT Y SUBGRADE 7080 PRINT

JOINT	INFLUENCE AREA
13	0.062500
14	0.062500
16	0.062500
18	0.031250
20	0.031250
21	0.062500
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162	0.062500
163	0.062500
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165	0.015625
166	0.031250
167	0.031250
168	0.031250
169	0.031250
170	0.031250
171	0.031250
172	0.031250
173	0.015625
174	0.031250
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176	0.031250
177	0.031250
178	0.031250
179	0.031250
180	0.031250
181	0.031250
182	0.015625

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    TOTAL          8.500000
 97. DEFINE MATERIAL START
 98. ISOTROPIC STEEL
 99. E 2.09042E+007
100. POISSON 0.3
101. DENSITY 7.83341
102. ALPHA 1.2E-005
103. DAMP 0.03
104. TYPE STEEL
105. STRENGTH FY 25819.2 FU 41584 RY 1.5 RT 1.2
106. ISOTROPIC CONCRETE
107. E 2.21467E+006
108. POISSON 0.17
109. DENSITY 2.40262
110. ALPHA 1E-005
111. DAMP 0.05
112. TYPE CONCRETE
113. STRENGTH FCU 2812.28
114. END DEFINE MATERIAL
115. MEMBER PROPERTY AMERICAN
116. 133 134 PRIS YD 0.55 ZD 0.55
117. CONSTANTS
118. MATERIAL CONCRETE ALL
119. LOAD 1 DL+LR
120. SELFWEIGHT Y -1
121. ELEMENT LOAD
122. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
123. JOINT LOAD
124. 145 FX 1.7 FY 9.54 FZ 0.04
125. 146 FX 2.18 FY 10.69 FZ -0.04
126. LOAD 2 DL+CL1
127. SELFWEIGHT Y -1
128. ELEMENT LOAD
129. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
130. JOINT LOAD
131. 145 FX 1.54 FY 9.46 FZ 0.04
132. 146 FX 1.62 FY 8.96 FZ -0.04
133. LOAD 3 DL+CL2
134. SELFWEIGHT Y -1
135. ELEMENT LOAD
136. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
137. JOINT LOAD
138. 146 FX -0.02 FY -28.79 FZ 2.88
139. 145 FX 0.45 FY -28.81 FZ -2.7
140. LOAD 4 DL+CL3
141. SELFWEIGHT Y -1
142. ELEMENT LOAD
143. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
144. JOINT LOAD
145. 145 FX 1.17 FY 9.11 FZ 0.04
146. 146 FX 1.25 FY 8.62 FZ -0.04
147. LOAD 5 DL+CL4
148. SELFWEIGHT Y -1
149. ELEMENT LOAD
150. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
151. JOINT LOAD
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152. 145 FX 1.56 FY 11.08 FZ 0.04
153. 146 FX 1.64 FY 10.59 FZ -0.04
154. LOAD 6 DL+CL5
155. SELFWEIGHT Y -1
156. ELEMENT LOAD
157. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
158. JOINT LOAD
159. 145 FX 1.15 FY 10.74 FZ 0.04
160. 146 FX 1.23 FY 10.24 FZ -0.04
161. LOAD 7 DL+CL6
162. SELFWEIGHT Y -1
163. ELEMENT LOAD
164. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
165. JOINT LOAD
166. 145 FX 1.36 FY 9.74 FZ 0.17
167. 146 FX 1.43 FY 8.34 FZ 0.09
168. LOAD 8 DL+CL5
169. SELFWEIGHT Y -1
170. ELEMENT LOAD
171. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
172. JOINT LOAD
173. 146 FX -0.05 FY -24.61 FZ 2.4
174. 145 FX 0.14 FY -25.6 FZ -2.41
175. LOAD 9 DL+CL6
176. SELFWEIGHT Y -1
177. ELEMENT LOAD
178. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
179. JOINT LOAD
180. 145 FX 1.35 FY 8.83 FZ -0.09
181. 146 FX 1.44 FY 9.24 FZ -0.17
182. LOAD 10 DL+CL7
183. SELFWEIGHT Y -1
184. ELEMENT LOAD
185. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
186. JOINT LOAD
187. 145 FX 1.36 FY 11.36 FZ 0.17
188. 146 FX 1.43 FY 9.97 FZ 0.09
189. LOAD 11 DL+CL8
190. SELFWEIGHT Y -1
191. ELEMENT LOAD
192. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
193. JOINT LOAD
194. 145 FX 1.07 FY 6.76 FZ -0.12
195. 146 FX 1.15 FY 7.35 FZ -0.2
196. LOAD 12 DL+0.75*CL1+0.75*LR
197. SELFWEIGHT Y -1
198. ELEMENT LOAD
199. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
200. JOINT LOAD
201. 145 FX 1.07 FY 6.76 FZ -0.12
202. 146 FX 1.15 FY 7.35 FZ -0.2
203. LOAD 13 DL+0.75*CL2+0.75*LR
204. SELFWEIGHT Y -1
205. ELEMENT LOAD
206. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
207. JOINT LOAD
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208. 145 FX 1.94 FY 10.66 FZ 0.04
209. 146 FX 2.33 FY 11.4 FZ -0.04
210. LOAD 14 DL+0.75*CL3+0.75*LR
211. SELFWEIGHT Y -1
212. ELEMENT LOAD
213. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
214. JOINT LOAD
215. 145 FX 1.96 FY 11.88 FZ 0.04
216. 146 FX 2.34 FY 12.62 FZ -0.04
217. LOAD 15 DL+0.75*CL4+0.75*LR
218. SELFWEIGHT Y -1
219. ELEMENT LOAD
220. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
221. JOINT LOAD
222. 145 FX 1.65 FY 11.62 FZ 0.04
223. 146 FX 2.04 FY 12.36 FZ -0.04
224. LOAD 16 DL+0.75*CL5+0.75*LR
225. ELEMENT LOAD
226. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
227. JOINT LOAD
228. 145 FX 1.81 FY 10.87 FZ 0.14
229. 146 FX 2.19 FY 10.93 FZ 0.06
230. SELFWEIGHT Y -1
231. LOAD 17 DL+0.75*CL6+0.75*LR
232. SELFWEIGHT Y -1
233. ELEMENT LOAD
234. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
235. JOINT LOAD
236. 145 FX 1.8 FY 10.19 FZ -0.06
237. 146 FX 2.19 FY 11.61 FZ -0.14
238. LOAD 18 DL+0.75*CL7+0.75*LR
239. JOINT LOAD
240. 145 FX 1.81 FY 12.09 FZ 0.14
241. 146 FX 2.19 FY 12.15 FZ 0.06
242. SELFWEIGHT Y -1
243. ELEMENT LOAD
244. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
245. LOAD 19 DL+0.75*CL8+0.75*LR
246. SELFWEIGHT Y -1
247. ELEMENT LOAD
248. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
249. JOINT LOAD
250. 145 FX 1.59 FY 8.64 FZ -0.08
251. 146 FX 1.97 FY 10.19 FZ -0.16
252. LOAD 20 DL+0.75*LR+0.75*WL1
253. JOINT LOAD
254. 145 FX 2.03 FY 8.49 FZ 0.17
255. 146 FX 2.22 FY 9.21 FZ 0.03
256. SELFWEIGHT Y -1
257. ELEMENT LOAD
258. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
259. LOAD 21 DL+0.75*LR+0.75*WL2
260. SELFWEIGHT Y -1
261. ELEMENT LOAD
262. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
263. JOINT LOAD
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264. 145 FX 2.03 FY 7.63 FZ 0.35
265. 146 FX 2.39 FY 8.37 FZ 0.11
266. LOAD 22 DL+0.75*LR+0.75*WL3
267. SELFWEIGHT Y -1
268. ELEMENT LOAD
269. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
270. JOINT LOAD
271. 145 FX 0.08 FY 7.29 FZ 0.16
272. 146 FX 0.47 FY 8.03 FZ 0.02
273. LOAD 23 DL+0.75*LR+0.75*WL4
274. SELFWEIGHT Y -1
275. ELEMENT LOAD
276. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
277. JOINT LOAD
278. 145 FX 0.23 FY 6.5 FZ 0.35
279. 146 FX 0.62 FY 7.27 FZ 0.1
280. LOAD 24 DL+0.75*LR+0.75*WL5
281. SELFWEIGHT Y -1
282. ELEMENT LOAD
283. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
284. JOINT LOAD
285. 145 FX 1.47 FY 9.78 FZ 0.73
286. 146 FX 1.84 FY 6.26 FZ 0.61
287. LOAD 25 DL+0.75*LR+0.75*WL6
288. SELFWEIGHT Y -1
289. ELEMENT LOAD
290. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
291. JOINT LOAD
292. 145 FX 1.6 FY 8.57 FZ 0.82
293. 146 FX 1.96 FY 5.53 FZ 0.62
294. LOAD 26 DL+0.75*LR+0.75*WL7
295. SELFWEIGHT Y -1
296. ELEMENT LOAD
297. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
298. JOINT LOAD
299. 145 FX 1.38 FY 5.36 FZ -0.91
300. 146 FX 1.78 FY 10.5 FZ -0.81
301. LOAD 27 DL+0.75*LR+0.75*WL8
302. SELFWEIGHT Y -1
303. ELEMENT LOAD
304. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.2
305. JOINT LOAD
306. 145 FX 1.56 FY 4.75 FZ -0.63
307. 146 FX 1.97 FY 9.34 FZ -0.64
308. LOAD 28 0.6*DL+WL1
309. SELFWEIGHT Y -1
310. ELEMENT LOAD
311. 25 TO 76 81 TO 132 135 TO 166 PR GY -3.3
312. JOINT LOAD
313. 145 FX 1.27 FY 3.7 FZ 0.2
314. 146 FX 1.06 FY 3.36 FZ 0.07
315. LOAD 29 0.6*DL+WL2
316. SELFWEIGHT Y -1
317. ELEMENT LOAD
318. 25 TO 76 81 TO 132 135 TO 166 PR GY -3.3
319. JOINT LOAD
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320. 146 FX -2.98 FY -17.23 FZ 1.53
321. 145 FX -2.82 FY -18.09 FZ -1.49
322. LOAD 30 0.6*DL+WL3
323. SELFWEIGHT Y -1
324. ELEMENT LOAD
325. 25 TO 76 81 TO 132 135 TO 166 PR GY -3.3
326. JOINT LOAD
327. 145 FX 1.27 FY 2.55 FZ 0.44
328. 146 FX 1.29 FY 2.25 FZ 0.17
329. LOAD 31 0.6*DL+WL4
330. SELFWEIGHT Y -1
331. ELEMENT LOAD
332. 25 TO 76 81 TO 132 135 TO 166 PR GY -3.3
333. JOINT LOAD
334. 145 FX -1.12 FY 1.05 FZ 0.43
335. 146 FX -1.08 FY 0.78 FZ 0.16
336. LOAD 32 0.6*DL+WL5
337. SELFWEIGHT Y -1
338. ELEMENT LOAD
339. 25 TO 76 81 TO 132 135 TO 166 PR GY -3.3
340. JOINT LOAD
341. 145 FX 0.53 FY 5.41 FZ 0.95
342. 146 FX 0.55 FY -0.56 FZ 0.84
343. LOAD 33 0.6*DL+WL6
344. SELFWEIGHT Y -1
345. ELEMENT LOAD
346. 25 TO 76 81 TO 132 135 TO 166 PR GY -3.3
347. JOINT LOAD
348. 145 FX 0.7 FY 3.8 FZ 1.07
349. 146 FX 0.71 FY -1.54 FZ 0.86
350. LOAD 34 0.6*DL+WL7
351. SELFWEIGHT Y -1
352. ELEMENT LOAD
353. 25 TO 76 81 TO 132 135 TO 166 PR GY -3.3
354. JOINT LOAD
355. 145 FX 0.41 FY -0.48 FZ -1.24
356. 146 FX 0.47 FY 5.09 FZ -1.05
357. LOAD 35 0.6*DL+WL8
358. SELFWEIGHT Y -1
359. ELEMENT LOAD
360. 25 TO 76 81 TO 132 135 TO 166 PR GY -3.3
361. JOINT LOAD
362. 145 FX 0.64 FY -1.3 FZ -0.87
363. 146 FX 0.72 FY 3.54 FZ -0.83
364. ************************************
365. LOAD 100 1.4*DL
366. SELFWEIGHT Y -1
367. ELEMENT LOAD
368. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.5
369. JOINT LOAD
370. 145 FX 1.36 FY 10.04 FZ 0.05
371. 146 FX 1.47 FY 9.35 FZ -0.05
372. LOAD 101 1.4D+1.7(CL1)+0.5LR
373. SELFWEIGHT Y -1
374. ELEMENT LOAD
375. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.5
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376. JOINT LOAD
377. 145 FX 2.69 FY 15.11 FZ 0.06
378. 146 FX 3.01 FY 15.24 FZ -0.05
379. LOAD 102 1.4D+1.7(CL2)+0.5LR
380. SELFWEIGHT Y -1
381. ELEMENT LOAD
382. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.5
383. JOINT LOAD
384. 145 FX 2.06 FY 14.52 FZ 0.06
385. 146 FX 2.37 FY 14.65 FZ -0.05
386. LOAD 103 1.4D+1.7(CL3)+0.5LR
387. SELFWEIGHT Y -1
388. ELEMENT LOAD
389. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.5
390. JOINT LOAD
391. 145 FX 2.73 FY 17.87 FZ 0.06
392. 146 FX 3.04 FY 18 FZ -0.05
393. LOAD 104 1.4D+1.7(CL4)+0.5LR
394. SELFWEIGHT Y -1
395. ELEMENT LOAD
396. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.5
397. JOINT LOAD
398. 145 FX 2.03 FY 17.28 FZ 0.06
399. 146 FX 2.34 FY 17.41 FZ -0.05
400. LOAD 105 1.4D+1.7(CL5)+0.5LR
401. SELFWEIGHT Y -1
402. ELEMENT LOAD
403. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.5
404. JOINT LOAD
405. 145 FX 2.38 FY 15.58 FZ 0.28
406. 146 FX 2.69 FY 14.18 FZ 0.17
407. LOAD 106 1.4D+1.7(CL6)+0.5LR
408. SELFWEIGHT Y -1
409. ELEMENT LOAD
410. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.5
411. JOINT LOAD
412. 145 FX 2.37 FY 14.05 FZ -0.17
413. 146 FX 2.69 FY 15.71 FZ -0.28
414. LOAD 107 1.4D+1.7(CL7)+0.5LR
415. SELFWEIGHT Y -1
416. ELEMENT LOAD
417. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.5
418. JOINT LOAD
419. 145 FX 2.38 FY 18.35 FZ 0.28
420. 146 FX 2.69 FY 16.94 FZ 0.17
421. LOAD 108 1.4D+1.7(CL8)+0.5LR
422. SELFWEIGHT Y -1
423. ELEMENT LOAD
424. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.5
425. JOINT LOAD
426. 145 FX 1.9 FY 10.53 FZ -0.21
427. 146 FX 2.21 FY 12.5 FZ -0.32
428. LOAD 109 1.2D+1.6LR+1.0(CL1)
429. SELFWEIGHT Y -1
430. ELEMENT LOAD
431. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.1
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432. JOINT LOAD
433. 145 FX 2.9 FY 14.68 FZ 0.05
434. 146 FX 3.64 FY 16.72 FZ -0.05
435. LOAD 110 1.2D+1.6LR+1.0(CL2)
436. SELFWEIGHT Y -1
437. ELEMENT LOAD
438. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.1
439. JOINT LOAD
440. 145 FX 2.53 FY 14.33 FZ 0.05
441. 146 FX 3.27 FY 16.37 FZ -0.05
442. LOAD 111 1.2D+1.6LR+1.0(CL3)
443. SELFWEIGHT Y -1
444. ELEMENT LOAD
445. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.1
446. JOINT LOAD
447. 145 FX 2.92 FY 16.3 FZ 0.05
448. 146 FX 3.66 FY 18.34 FZ -0.05
449. LOAD 112 1.2D+1.6LR+1.0(CL4)
450. SELFWEIGHT Y -1
451. ELEMENT LOAD
452. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.1
453. JOINT LOAD
454. 145 FX 2.51 FY 15.96 FZ 0.05
455. 146 FX 3.25 FY 18 FZ -0.05
456. LOAD 113 1.2D+1.6LR+1.0(CL5)
457. SELFWEIGHT Y -1
458. ELEMENT LOAD
459. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.1
460. JOINT LOAD
461. 145 FX 2.71 FY 14.96 FZ 0.18
462. 146 FX 3.46 FY 16.1 FZ 0.08
463. LOAD 114 1.2D+1.6LR+1.0(CL6)
464. SELFWEIGHT Y -1
465. ELEMENT LOAD
466. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.1
467. JOINT LOAD
468. 145 FX 2.71 FY 14.05 FZ -0.08
469. 146 FX 3.46 FY 17 FZ -0.18
470. LOAD 115 1.2D+1.6LR+1.0(CL7)
471. SELFWEIGHT Y -1
472. ELEMENT LOAD
473. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.1
474. JOINT LOAD
475. 145 FX 2.71 FY 16.58 FZ 0.18
476. 146 FX 3.46 FY 17.72 FZ 0.08
477. LOAD 116 1.2D+1.6LR+1.0(CL8)
478. SELFWEIGHT Y -1
479. ELEMENT LOAD
480. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.1
481. JOINT LOAD
482. 145 FX 2.43 FY 11.98 FZ -0.11
483. 146 FX 3.17 FY 15.11 FZ -0.2
484. LOAD 117 1.2D+1.6WL1+1.0L+0.5LR
485. SELFWEIGHT Y -1
486. ELEMENT LOAD
487. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.1
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488. JOINT LOAD
489. 145 FX 2.62 FY 8.83 FZ 0.33
490. 146 FX 2.51 FY 8.99 FZ 0.1
491. LOAD 118 1.2D+1.6WL2+1.0L+0.5LR
492. SELFWEIGHT Y -1
493. ELEMENT LOAD
494. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.1
495. JOINT LOAD
496. 145 FX 2.63 FY 6.99 FZ 0.71
497. 146 FX 2.88 FY 7.21 FZ 0.26
498. LOAD 119 1.2D+1.6WL3+1.0L+0.5LR
499. SELFWEIGHT Y -1
500. ELEMENT LOAD
501. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.1
502. JOINT LOAD
503. 145 FX -1.53 FY 6.25 FZ 0.31
504. 146 FX -1.23 FY 6.48 FZ 0.08
505. LOAD 120 1.2D+1.6WL4+1.0L+0.5LR
506. SELFWEIGHT Y -1
507. ELEMENT LOAD
508. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.1
509. JOINT LOAD
510. 145 FX -1.2 FY 4.58 FZ 0.7
511. 146 FX -0.9 FY 4.85 FZ 0.25
512. LOAD 121 1.2D+1.6WL5+1.0L+0.5LR
513. SELFWEIGHT Y -1
514. ELEMENT LOAD
515. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.1
516. JOINT LOAD
517. 145 FX 1.44 FY 11.56 FZ 1.52
518. 146 FX 1.71 FY 2.71 FZ 1.34
519. LOAD 122 1.2D+1.6WL6+1.0L+0.5LR
520. SELFWEIGHT Y -1
521. ELEMENT LOAD
522. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.1
523. JOINT LOAD
524. 145 FX 1.71 FY 8.99 FZ 1.72
525. 146 FX 1.96 FY 1.14 FZ 1.36
526. LOAD 123 1.2D+1.6WL7+1.0L+0.5LR
527. SELFWEIGHT Y -1
528. ELEMENT LOAD
529. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.1
530. JOINT LOAD
531. 145 FX 1.25 FY 2.14 FZ -1.97
532. 146 FX 1.57 FY 11.76 FZ -1.7
533. LOAD 124 1.2D+1.6WL8+1.0L+0.5LR
534. SELFWEIGHT Y -1
535. ELEMENT LOAD
536. 25 TO 76 81 TO 132 135 TO 166 PR GY -5.1
537. JOINT LOAD
538. 145 FX 1.63 FY 0.83 FZ -1.39
539. 146 FX 1.97 FY 9.28 FZ -1.33
540. LOAD 125 0.9D+1.6WL1
541. SELFWEIGHT Y -1
542. ELEMENT LOAD
543. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.5
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544. JOINT LOAD
545. 145 FX 1.97 FY 5.49 FZ 0.32
546. 146 FX 1.63 FY 4.98 FZ 0.11
547. LOAD 126 0.9D+1.6WL2
548. SELFWEIGHT Y -1
549. ELEMENT LOAD
550. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.5
551. JOINT LOAD
552. 145 FX 1.97 FY 3.65 FZ 0.7
553. 146 FX 2 FY 3.2 FZ 0.27
554. LOAD 127 0.9D+1.6WL3
555. SELFWEIGHT Y -1
556. ELEMENT LOAD
557. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.5
558. JOINT LOAD
559. 145 FX -2.18 FY 2.92 FZ 0.3
560. 146 FX -2.11 FY 2.47 FZ 0.09
561. LOAD 128 0.9D+1.6WL4
562. SELFWEIGHT Y -1
563. ELEMENT LOAD
564. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.5
565. JOINT LOAD
566. 145 FX -1.86 FY 1.24 FZ 0.69
567. 146 FX -1.79 FY 0.84 FZ 0.26
568. LOAD 129 0.9D+1.6WL5
569. SELFWEIGHT Y -1
570. ELEMENT LOAD
571. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.5
572. JOINT LOAD
573. 145 FX 0.78 FY 8.23 FZ 1.51
574. 146 FX 0.82 FY -1.3 FZ 1.35
575. LOAD 130 0.9D+1.6WL6
576. SELFWEIGHT Y -1
577. ELEMENT LOAD
578. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.5
579. JOINT LOAD
580. 145 FX 1.06 FY 5.65 FZ 1.71
581. 146 FX 1.08 FY -2.87 FZ 1.37
582. LOAD 131 0.9D+1.6WL7
583. SELFWEIGHT Y -1
584. ELEMENT LOAD
585. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.5
586. JOINT LOAD
587. 145 FX 0.6 FY -1.19 FZ -1.99
588. 146 FX 0.69 FY 7.75 FZ -1.69
589. LOAD 132 0.9D+1.6WL8
590. SELFWEIGHT Y -1
591. ELEMENT LOAD
592. 25 TO 76 81 TO 132 135 TO 166 PR GY -4.5
593. JOINT LOAD
594. 145 FX 0.97 FY -2.51 FZ -1.4
595. 146 FX 1.09 FY 5.27 FZ -1.32
596. PERFORM ANALYSIS
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P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 164 NUMBER OF MEMBERS 2
NUMBER OF PLATES 136 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 162

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 142/ 15/ 48 DOF
TOTAL PRIMARY LOAD CASES = 68, TOTAL DEGREES OF FREEDOM = 498
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 24 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 13.4/ 111422.4 MB

597. LOAD LIST 100 TO 109
598. START CONCRETE DESIGN

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599. CODE ACI 1999
600. TRACK 0 ALL
601. MAXMAIN 25 ALL
602. CLB 0.075 MEMB 133 134
603. CLS 0.075 ALL
604. DESIGN COLUMN 133 134
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COLUMN NO. 133 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	100	END	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

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COLUMN NO. 134 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	100	END	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

**********END OF COLUMN DESIGN RESULTS**********

605. END CONCRETE DESIGN
606. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= JUN 5,2020 TIME= 11:25:15 ****

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* *
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